

The Future is Agentic: Definitions, Perspectives, and Open Challenges of Multi-Agent Recommender Systems

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Large language models (LLMs) are evolving from passive text generators into agentic systems that can plan, maintain state, invoke tools, and coordinate with other agents. This perspective paper examines what this shift means for recommender systems (RS). We define *agentic recommender systems* as recommendation pipelines in which one or more stateful agents observe, plan, call tools, and verify, rather than score in a single shot, while operating over users, item catalogs, candidate sets, and recommendation objectives. Their value is strongest when this added machinery measurably improves recommendation-layer outcomes such as relevance, constraint satisfaction, bundle coherence, grounding, explanation faithfulness, fairness of exposure, or user effort, and is not justified merely because a pipeline contains an LLM or several modules. To make these notions precise for recommendation rather than for agents in general, we introduce a recommender-specific formalism that models an individual recommender agent by its state (user, context, history, and candidate set), a reasoning core, recommendation-specific tools, a hierarchical memory, and explicit policy constraints, and captures a multi-agent recommender as a triple of agents, a shared environment exposing the item catalog and feedback signals, and a communication protocol. Within this framework, we develop four representative task families (interactive goal-oriented recommendation, user simulation and evaluation, contextual and multimodal recommendation, and grounded explanation) and an operational agenda that ties five recurring challenge families (communication protocols, scalability and cost, hallucination and error propagation, emergent misalignment and collusion, and brand and policy compliance) to measurable RS signals. Finally, we conduct a controlled empirical study comparing single-shot and multi-agent pipelines under shared user histories, candidate sets, prompts, and metrics. A pilot next-item ranking study on Amazon-2023 categories shows that multi-agent systems are not uniformly superior: on representative samples the single-shot baseline is Pareto-efficient, whereas decomposition and ensemble agents become useful mainly for high-diversity user histories. This supports a conditional design principle: agentic complexity should be routed to the cases where its marginal quality improvement justifies the additional latency, cost, and governance risk. To support reproduction, we release all pipeline implementations, prompts, and results here: <https://github.com/RezaYM/agenticrecsys.git>.

CCS Concepts: • **Information systems** → **Recommender systems**.

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1 Introduction and Motivation

Large Language Model (LLM) agents go beyond traditional chatbots by offering goal-directed, agentic behavior rather than merely responding to user queries through one-shot text generation.

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In essence, they are designed to handle multi-step tasks, orchestrate information flow, and autonomously employ tools or functions when necessary [48, 61, 84]. This distinction means that while a conventional chatbot might provide short answers in a single round of dialogue, an agentic system can proactively structure a complex problem and solve it through a sequence of methodical steps. Put another way, an LLM agent is not only a reactive conversational partner but a dynamic problem-solver capable of decomposing tasks, maintaining state, invoking external resources, and adapting its strategy to reach a goal [64, 66, 76].

Much as goal-directed agentic behavior has reshaped other NLP-oriented tasks, it applies naturally to the tasks surrounding recommender systems (RS). In the most common case, recommendation is handled in a single shot: given a user history and a candidate set, a retrieval or ranking function estimates relevance and returns a ranked list of similar movies or items. Agentic recommendation broadens this picture along two axes. First, even a single-shot task can sometimes be handled better by decomposing it into subtasks, for example by summarizing a long or heterogeneous history, planning a ranking strategy, or aggregating several candidate orderings before committing to a final list. Second, and more fundamentally, the user need may not be “*rank items for this profile*” at all, but rather “*help me achieve a goal*” under multiple constraints, evolving context, and incomplete information. In this latter regime, recommendation becomes a multi-stage decision process rather than a single scoring operation: the system may ask clarifying questions, retrieve current catalog evidence, reason over constraints, call external tools, maintain memory over prior interactions, verify whether candidate items satisfy user and policy constraints, and finally produce a ranked list, bundle, explanation, or action plan. Section 4 develops these possibilities through four representative scenarios: *interactive goal-oriented recommendation*, *user simulation for offline evaluation*, *contextual and multimodal recommendation*, and *recommendation explanation*.

We refer to this RS-centered paradigm as *agentic recommendation*. In an agentic recommender system, recommendation is no longer only a one-shot call to a ranker; it is a goal-driven process in which one or more stateful agents participate in an observe–plan–act–verify loop over user signals, item evidence, tools, memories, constraints, and recommendation objectives. This viewpoint gives us the flexibility, and the obligation, to evaluate recommendation along a more diverse set of metrics than top- K accuracy alone: relevance and ranking quality, constraint satisfaction, diversity, bundle coherence, evidence grounding, explanation faithfulness, fairness of exposure, long-term user value, and user effort, together with trace-level signals such as tool-call success, memory correctness, latency, and cost. The agentic layer is useful when it improves these recommendation-relevant outcomes, not merely because a system contains an LLM or several modules.

Beyond the multi-step task handling already described, agentic recommendation rests on two further capabilities in particular: **memory**, so that the agent can carry state across the steps of a task, and **tool use**, so that it can act on evidence beyond its parameters.

The first, **memory**, is a core mechanism for agentic recommendation rather than an optional implementation detail. Over a multi-step task the system may need to remember the last item shown in a session, a rejected style, a durable preference inferred across interactions, or a recurring workflow that should be executed with minimal friction. We use the standard distinction between *working*, *episodic*, *semantic*, and *procedural* memory, and defer the taxonomy, storage mechanisms, and update/retrieval operators to Section 3. What matters at this point is that memory is not only an enabler but also an object of evaluation: stale memories can contaminate future rankings, incorrect summaries can distort user profiles, and privacy-sensitive traces may be retrieved when they should be forgotten. Memory should therefore be judged by whether it improves downstream recommendation utility, preserves relevant constraints, avoids stale or irrelevant facts, and respects privacy and deletion requirements.

The second, *autonomous tool use*, lets the agent act on evidence beyond its parameters. Instead of relying only on static model parameters, agentic recommenders can invoke retrieval APIs, product databases, image-analysis tools, policy checkers, constraint solvers, or evaluation modules to fetch data, analyze content, and act on domain-specific evidence [27, 57]. For example, a request for a restaurant can trigger queries for current ratings, availability, location, and dietary filters rather than depending on memorized information; a request to furnish an uploaded room image can trigger a vision tool to extract visual features and then a catalog query for items matching the user's style and spatial constraints. Tool use and memory are complementary: memory carries durable preferences and prior recommendations across steps, while tools ground the next recommendation in current catalog, policy, or contextual evidence.

The framework developed in this paper is therefore LLM-centric but not LLM-exclusive. LLMs are currently a natural implementation substrate for agentic recommendation because they provide flexible natural-language understanding, reasoning, tool-use interfaces, and multi-agent communication. However, an agent in an agentic recommender may also be powered by a classical ranker, a retrieval model, a smaller language model, a rule-based controller, a constraint solver, a vision model, a simulator, or a human-review module. What makes the system recommender-specific is not the mere use of natural language or multiple agents, but the fact that these components operate over users, contexts, histories, item catalogs, candidate sets, ranking objectives, feedback signals, explanations, and exposure effects.

Taken together, (i) multi-step task handling, (ii) memory retention, and (iii) autonomous tool use empower LLM-based and non-LLM-based agents to operate with a level of autonomy that transcends the simpler question-and-answer paradigm of traditional chatbots [48, 61, 64]. At the same time, this should not be read as a claim that agentic systems are always preferable. For stable, well-scoped, high-throughput ranking tasks, conventional recommender pipelines may remain simpler, cheaper, more predictable, and easier to evaluate. The value of agentic recommendation is therefore strongest when planning, memory, tool use, interaction, and verification improve recommendation-layer outcomes enough to justify their additional latency, cost, privacy risk, and governance complexity.

1.1 A Motivating Example: Goal-Oriented Birthday-Party Recommendation

Figure 1 illustrates the type of recommendation task that motivates an agentic design. The user is not asking for a single ranked list of products from a static profile. Instead, the user begins with an open-ended goal: "a Mickey Mouse-themed birthday party." The recommender then elicits missing constraints by asking about color preferences and dietary restrictions. When the user answers "Gluten free," a dietary constraint is introduced that must be respected by the recommended collection. After the system proposes an initial set of party items, the user further revises the goal by asking whether the cake can be made larger and whether balloons can be added. This interaction shows that the task is not merely to rank products, but to support an evolving, goal-oriented recommendation process: construct a coherent themed birthday-party bundle, satisfy dietary restrictions, adapt item attributes based on feedback, and add new item categories as the conversation unfolds.

To handle this complex natural-language intent, the system decomposes the broad goal into recommendation-relevant subtasks:

- A user-context component retrieves session-level and long-term traces, including recent dialogue state, prior preferences, accepted and rejected recommendations, and relevant constraints.

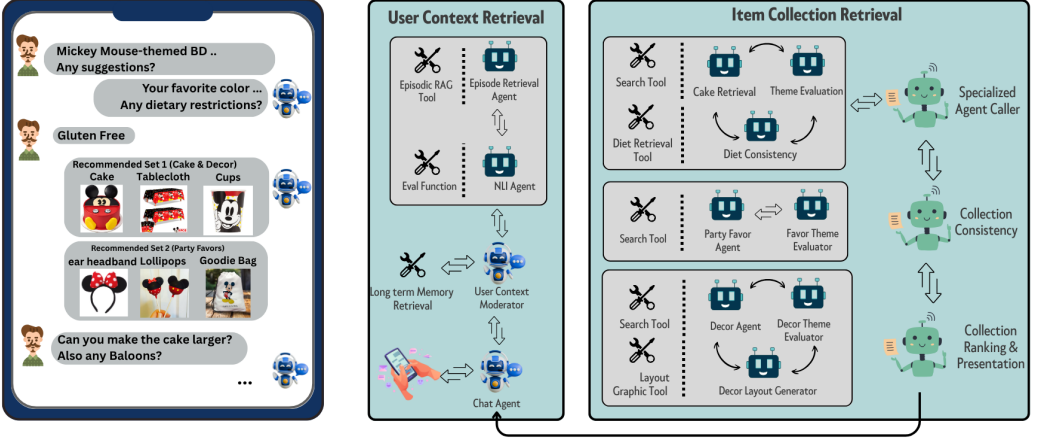


Fig. 1. Left: Conversation example for a goal-oriented collection recommendation, where the user specifies an open-ended goal instead of searching for specific items. Right: Architecture for personalized goal-oriented recommendation using a multi-agent pipeline with specialized agents and tools.

- A specialized-agent caller dynamically instantiates sub-agents equipped with retrieval or generation tools for distinct recommendation categories, such as cakes, decorations, favors, or layout suggestions.
- Category-specific agents retrieve candidate items from catalog or search tools while respecting explicit constraints such as theme, dietary requirements, budget, and availability.
- A collection-consistency agent checks whether the resulting items cohere as a bundle, for example whether the cake, tablecloth, decor, and party favors follow the same theme and do not violate constraints.
- A ranking and presentation agent orders the curated item set according to user preferences, contextual priorities, and recommendation objectives, and can generate a user-facing explanation or visual layout.

Throughout this interaction, memory carries the evolving recommendation state across turns: the theme, the gluten-free constraint, the accepted and rejected items, and the later requests to enlarge the cake and add balloons. The example also makes the evaluation concern concrete, since the system must decide which of these memories are still relevant, which are stale, and which should be excluded for privacy, safety, or correctness reasons.

1.2 Organization of the Paper and Contributions

This is a perspective and conceptual framework paper, not a new learning algorithm and not an exhaustive survey. Its aim is to give the recommender-systems community a shared vocabulary and an evaluation-oriented agenda for agentic RS: precise definitions of agents, memory, tools, actions, communication protocols, traces, objectives, and evaluation signals, so that future work can describe architectures, compare systems, and separate recommender-specific contributions from general agent engineering. Throughout, we take the position that the value of agentic recommendation is strongest when orchestration and interaction measurably improve recommendation-layer outcomes, rather than when a pipeline merely adds LLM calls or agent coordination. To keep this distinction sharp, we separate a recommendation layer (candidate generation, ranking, and slate

construction), an interaction layer (dialogue, clarification, explanation, and preference elicitation), and an orchestration layer (tool choice, agent coordination, memory management, and verification), defined in Section 2.5.

The paper makes four contributions.

- **Core vocabulary, formal framework, and scope boundary (Section 2).** We give the minimal architectural primitives needed to specify an agentic recommender system: recommender agents, user and item state, candidate sets, memory stores, tool interfaces, action spaces, communication protocols, constraints, objectives, and observable traces. We distinguish generic LLM agents from recommender agents, and separate recommendation-layer contributions from interaction-layer capabilities and generic orchestration infrastructure.
- **Running example and task families (Fig. 1; Sections 3–4).** Using the Mickey Mouse birthday-planning example as a running case, we show how specialized agents, toolchains, memory hierarchies, and verification modules compose for goal-oriented recommendation. We then develop four representative task families, interactive recommendation, user simulation and evaluation, contextual and multimodal recommendation, and grounded recommendation explanation, and indicate where a conventional ranking pipeline remains simpler, cheaper, and more reliable.
- **Operational challenge and evaluation agenda (Section 5).** We organize the open problems of agentic recommendation into five challenge families, communication complexity and protocol design, scalability and cost, hallucination and error propagation, emergent misalignment and collusion, and brand, policy, and legal compliance, and connect each to measurable recommender-system signals such as ranking quality, evidence coverage, trace validity, latency, token cost, exposure disparity, policy-violation rate, and judge reliability.
- **Controlled empirical study (Section 6).** We report a controlled experiment comparing single-shot LLM call and multi-agent recommendation pipelines under shared user histories, candidate sets, prompts, and metrics. A pilot next-item ranking study shows that multi-agent pipelines are conditionally useful: they do not dominate on representative samples, but decomposition and ensemble roles help on high-diversity histories.

2 Agentic Recommender Systems: Definition, Characteristics, Misconceptions

In this section, we clarify the concept of an *LLM agent*, in more depth. When illustrating through examples we use the context of recommender and retrieval systems. We also discuss common misconceptions about agentic pipelines and elaborate on what is *not* an LLM agent.

2.1 What Defines an LLM Agent

An **LLM agent**, in its fully functional capacity, is an AI system in which a large language model (LLM) serves as the core decision-making component (the “brain”), enhanced by additional mechanisms that enable it to carry out complex, multi-step tasks autonomously rather than relying on a single prompt-response. In other words, this LLM is part of a larger architecture that provides (i) planning abilities, (ii) memory, (iii) tool/API usage, for interactions with external systems, (iv) an *autonomous decision loop* that can break down a goal, observe intermediate steps, and adapt its strategy [61, 84]. In a broader context, multiple such language agents form a **multi-agent system (MAS)**, in which agents can interact, communicate, coordinate, and even compete—leveraging their reasoning and communication capabilities to collectively solve complex tasks that exceed the capacity of any single agent.

This design (i.e., agent-based approach) goes beyond the traditional static use of LLMs for single-turn queries and is increasingly used in recommendation and retrieval tasks. For example, an

Table 1. Core Capabilities of Agentic Recommender Systems

Icon	Capability	Description
	Planning & Task Decomposition	Breaks complex recommendation goals into sub-tasks, such as candidate selection, constraint checking, bundle retrieval, re-ranking, and verification.
	Tool Use & Action Execution	Invokes external tools or APIs, such as retrieval systems, inventory databases, policy checkers, vision models, or constraint solvers.
	Memory & State Management	Maintains state across turns and, when appropriate, across sessions using working, episodic, semantic, or procedural memory.
	Autonomy & Goal-Driven Behavior	Operates in a closed loop by observing context, choosing actions, evaluating intermediate outcomes, and revising the plan until the recommendation objective is satisfied or no further useful action is available.

LLM-based agent can iteratively search user logs to gather information about past item interactions before making tailored recommendations [58, 91].

2.2 Key Characteristics of LLM Agents

As summarized in Table 1, agentic recommender systems typically exhibit four core characteristics, described in the following:

- (1) **Planning and Task Decomposition:** The agent can execute (or even formulate) a plan. This is done by breaking a complex goal into subtasks. Rather than producing an immediate, one-shot answer, an agentic system can plan for and conduct a sequence of steps. This can involve reasoning to tackle long-horizon tasks [16, 54]. For example, in a recommender scenario, the planning module may first identify candidate items, then check user history, retrieve a consistent bundle of products, then re-rank items before generating a final recommendation [87].
- (2) **Tool Use and Action Execution:** An LLM agent is not limited to purely text-based outputs; it can *invoke external tools or APIs* and perform *actions* in an environment [84].¹ For instance, a recommendation agent could consult a real-time inventory database to see if recommended items are in stock, or it might retrieve user reviews from a knowledge base before finalizing its recommendations.
- (3) **Memory and State Management:** LLM agents maintain a notion of “state” across multiple steps. They often incorporate a memory module which can potentially store conversation context, past recommendations, user feedback, local guidelines, procedures, domain knowledge etc [82]. This memory can be in semantic [59], vector database [38], knowledge graph [4, 62] formats. The memory module is used by the agent so that it can recall relevant information at each step of executing a task. For instance, in recommendation scenarios, user preferences collected over multiple sessions can be stored in a memory module and retrieved for the task of personalizing the recommendations. This persistent memory is crucial for multi-turn recommendation dialogues, where the agent refines suggestions based on evolving user interests. [42, 80]
- (4) **Autonomy and Goal-Driven Behavior:** LLM agent can be designed to be operated in “autonomously in a closed-loop fashion” to fulfill a goal. Given a target objective (for example

¹Tool invocation examples include web searches, database queries, or calling a specialized retrieval API to filter items.

“find a suitable product for a user”), the agent can be designed to autonomously (i) observe the environment (via tools or APIs), (ii) evaluate progress, (iii) evaluate the outcome of each step and do a self refine step (iv) continue until it deems the goal is achieved or no further action can be taken. This is unlike a static Q&A system, where the system merely focuses on the one-time query. [52, 72]

These components enable the LLM agent to tackle complex tasks in recommendation and retrieval settings. For instance, systems like *RecMind* [76] use LLM as a reasoning engine paired with a planning component, a long-term memory of user profiles, and retrieval tools to fetch relevant product information. The agent then autonomously analyzes a user’s needs and iteratively refines its suggestions. This is far more capable than a single-turn Q&A approach, as it can plan queries (e.g., to check product ratings or availability), incorporate user feedback, and adapt the recommendation strategy. Frameworks like “ReAct” [84] illustrate how an LLM can be prompted to alternate between “reasoning steps” and “action steps”, calling retrieval or recommendation APIs as needed. Other systems, such as “Toolformer” [64] and “HuggingGPT” [66] similarly integrate LLMs with external action interfaces.

2.3 Formal Framework

In this subsection, we introduce formal definitions for key concepts central to our framework. These definitions establish a rigorous foundation for discussing the architectural and operational aspects of agents, including specialized LLM Agents, Multi-Agent Systems, and Tool-Using Agents. We also provide a generic formalism for representing the internal structure of an agent.

Definition 2.1 (LLM Agent). An *LLM Agent* is an intelligent system whose core decision-making and interaction capabilities are powered by one or more large language models. Formally, we denote such an agent as:

$$A_{\text{LLM}} = (\mathcal{M}, \mathcal{I}, \mathcal{O}, \mathcal{F}, \Omega),$$

where:

- $\mathcal{M}$ is the underlying language model or set of models used for text generation, understanding, and reasoning.
- $\mathcal{I}$ represents the input space that the agent can observe.
- $\mathcal{O}$ denotes the output space the agent can produce.
- $\mathcal{F}$ is a set of functions, tools, or APIs the agent can invoke to enhance its capabilities.
- Ω comprises any additional state or memory structures that enable the agent to maintain context across time or tasks.

For example, in recommendation settings, for an eCommerce shopping assistant, $\mathcal{M}$ can be a GPT-style transformer fine-tuned for e-commerce dialogue; it encodes linguistic knowledge and basic reasoning skills, allowing the agent to parse user requests and generate coherent replies. $\mathcal{I}$ is the space of user queries and contextual signals, while $\mathcal{O}$ is the space of possible outputs, including text responses, actions, or function calls. $\mathcal{F}$ represents external functions or APIs, such as retrieval APIs, price fetchers, or inventory status calls, which augment the innate knowledge of $\mathcal{M}$ with fresh, authoritative data, thereby mitigating hallucination. The agent’s memory, Ω , is partitioned into (a) *short-term memory* Ω^{STM} that keeps the last few dialogue turns, (b) *long-term episodic memory* Ω^{EPI} that stores past party-planning sessions, and (c) *semantic memory* Ω^{SEM} containing persistent user traits (e.g., “prefers red decor”). During inference, salient fragments of Ω are retrieved and appended to the prompt, giving the agent continuity across sessions.

For a multi-turn dialogue agent, at each timestep t , the agent receives an input $i_t \in \mathcal{I}$, consults its memory $\omega \subseteq \Omega$ to form an augmented context, and computes the output $o_t \in \mathcal{O}$ as follows:

$$o_t = f_{\mathcal{M}, \mathcal{F}}(i_t, \omega) \quad (1)$$

where $f_{\mathcal{M}, \mathcal{F}}$ is the agent's policy function that maps the current input i_t and retrieved memory ω to an output o_t , leveraging both the language model $\mathcal{M}$ and the available external functions $\mathcal{F}$.

Definition 2.1 intentionally describes a *generic* LLM agent. The same tuple becomes a *recommender agent* once each component is instantiated over recommendation-relevant objects rather than arbitrary text. Concretely, we take the agent to operate on a state

$$s_t = (u, C_t, H_u, \mathcal{S}_t^{(K)}, \pi),$$

where u identifies the user, $C_t \in \mathcal{I}$ is the dialogue or contextual state, H_u is a chronological interaction history, $\mathcal{S}_t^{(K)} \subseteq \mathcal{S}$ is the candidate set available at step t , and π encodes business, safety, privacy, and multi-stakeholder policy constraints. An LLM agent is a recommender agent when (i) its output space $\mathcal{O}$ contains recommendation actions such as `retrieve( $\cdot$ )`, `rank( $\cdot$ )`, `recommend( $\cdot$ )`, `ask( $\cdot$ )`, and `explain( $\cdot$ )`; (ii) its tool set $\mathcal{F}$ and environment expose RS-relevant evidence such as the item catalog, candidate sets, and feedback signals; and (iii) its objective is a recommendation utility (relevance, constraint satisfaction, long-term engagement) optimized under the constraints in π .

Definition 2.2 (Multi-Agent System (MAS)). A *Multi-Agent System* is an ordered triple

$$\text{MAS} = (\mathcal{A}, \mathcal{E}, \Pi),$$

where

- (a) $\mathcal{A} = \{A_1, \dots, A_n\}$ is a finite set of agents. Each A_i may be an LLM agent (Definition 2.1) or another kind of modular service (e.g., rule-based, vision, or database stub).
- (b) $\mathcal{E}$ is the shared *environment* that supplies percepts and resources to the agents—such as external APIs, user interfaces, or a simulated world state. Formally, $\mathcal{E}$ can be viewed as a partial observable Markov space from which each agent receives observations and in which it executes actions.
- (c) $\Pi = (\mathbf{C}, \Gamma)$ is the *interaction protocol*.
 - (i) $\mathbf{C} \in \{0, 1\}^{n \times n}$ is a (possibly time-varying) *communication matrix*. A value $C_{ij} = 1$ indicates that messages of type $\gamma \in \Gamma$ are *permitted* from A_i to A_j under current conditions. The condition can be
 - **preset** (static design-time routing), or
 - **autonomous** (dynamically toggled by agents. e.g., A_i opens a channel to A_j when cooperation is beneficial, closes it otherwise).
 - (ii) Γ is the finite set of admissible *message schemata* (performative types, serialization rules, timing constraints). Each $\gamma \in \Gamma$ defines both syntax and semantics so that a receiving agent can parse and act on the message.

An execution of MAS can be viewed as a sequence of environment states and inter-agent messages that respect Π . Depending on design, agents may coordinate (*co-operative*), compete (*self-interested*), or exhibit mixed motives while attempting to satisfy individual or shared objectives.

In the recommendation setting, the MAS triple specializes as follows. The environment $\mathcal{E}$ exposes recommendation state, including the item catalog, per-user histories, candidate sets, and observed feedback, rather than a generic world model. The agents in $\mathcal{A}$ occupy recommendation roles, such as profilers, retrievers, rankers, consistency checkers, evaluators, or user simulators, each a recommender agent in the sense above. The protocol Π carries recommendation payloads: messages

in Γ transport candidate lists, ranked slates, retrieved evidence, constraint or compliance reports, and memory records, and the communication matrix C encodes which of these exchanges are permitted. The marketplace example that follows is the smallest such instance.

Consider a minimal marketplace scenario with two autonomous components, namely a *recommendation agent* and an *evaluation agent*, so that

$$\mathcal{A} = \{A_{\text{rec}}, A_{\text{eval}}\}.$$

In this case $\mathcal{E} = (\text{UserProfileDB}, \text{ProductCatalogue}, \text{BusinessRulesKB})$, providing user attributes, real-time inventory, and brand-policy facts, respectively. The interaction protocol can be

$$\Pi = (C, \Gamma), \quad C = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \Gamma = \{\text{candidate_list}, \text{compliance_report}\}.$$

Thus A_{rec} may send a *candidate_list*, a ranked JSON array $\{(s_1, \text{score}_1), \dots, (s_L, \text{score}_L)\}$ to A_{eval} , which in turn replies with a *compliance_report* indicating any items that violate policy. The off-diagonal ones in C permit bidirectional messaging, while the zeros on the diagonal denote that agents do not address themselves. At runtime, $C_{\text{eval,rec}}$ can be toggled off once no further corrections are required, illustrating how links may be preset yet autonomously reconfigured according to system state.

Definition 2.3 (Memory Update Function). Let Ω_t denote the long-term memory state maintained by an agent at dialogue step t and let

$$C_t \in \mathcal{X}$$

be the raw context collected during that step (e.g., the most recent user utterance, the agent's reply, and any intermediate reasoning trace, represented in token or embedding space). A *Memory Update Function* is a mapping

$$\mathcal{U} : (C_t, \Omega_t) \longrightarrow \Omega_{t+1},$$

where

- $\mathcal{U}$ internally applies a *retention operator*

$$\mathcal{R} : C_t \rightarrow \tilde{C}_t$$

that distills C_t into a noise-reduced summary $\tilde{C}_t \in \tilde{\mathcal{X}}$,

- merges $\tilde{C}_t$ with the prior state Ω_t via a domain-specific *merge rule* $\diamond$, and produces the next persistent state:

$$\Omega_{t+1} = \Omega_t \diamond \tilde{C}_t.$$

The goal is to preserve salient semantic information while discarding redundancy, ensuring that Ω_{t+1} remains compact yet sufficient for future retrieval.

Assume a shopping assistant agent and a user that uses the agent to order a cake. At turn t the user says “remember that two of my guests require a gluten-free diet”. The raw context C_t includes the user utterance, the agent's internal chain-of-thought, and the pending action call to a *CakeSearch* tool. The retention operator extracts the key fact

$$\tilde{C}_t = \{\text{guest_allergy: gluten}\},$$

and the merge rule appends it to the agent's long-term *episodic* slot in Ω_t :

$$\Omega_{t+1}^{\text{EPI}} = \Omega_t^{\text{EPI}} \cup \{\text{guest_allergy: gluten}\}.$$

On the next turn ($t+1$) the planner retrieves this memory fragment and instructs a downstream cake-selection agent to filter for gluten-free options, illustrating how $\mathcal{U}$ enables continuity and correctness across dialogue turns without bloating the prompt window.

In a recommender, the update function $\mathcal{U}$ governs what persists about a user across turns and sessions: durable preferences, accepted and rejected items, satisfied or violated constraints, and session-level intent. The recsys-specific consideration is that $\mathcal{U}$ is not only an enabler of personalization but also a source of error. A retention operator $\mathcal{R}$ that over-summarizes can drop a binding constraint such as the gluten restriction; one that under-summarizes lets stale or contradictory preferences accumulate and later contaminate ranking. The merge rule $\diamond$ must therefore respect recommendation-relevant properties, namely recency, provenance, and the right to delete privacy-sensitive traces, so that Ω_{t+1} improves rather than degrades downstream recommendation utility. These properties are what the memory-correctness and staleness signals of Section 5 are intended to measure.

Definition 2.4 (Memory Retrieval Function). Let Ω be the agent’s persistent memory store after t dialogue steps. A *Memory Retrieval Function* is a mapping

$$Q : (\Omega, \tau) \longrightarrow \widehat{C},$$

where

- $\Omega = \{\widetilde{C}_1, \dots, \widetilde{C}_m\}$ is a corpus of distilled memory traces, each $\widetilde{C}_i \in \widetilde{\mathcal{X}}$ (text snippets, embeddings, or hybrids) produced by the update operator $\mathcal{U}$ (Definition 2.3).
- $\tau \in \mathcal{T}$ is a task representation—e.g. the current user query, an intermediate reasoning goal, or a structured tool call argument—expressed in text, vector, or multi-modal form.
- $\widehat{C} \in \widehat{\mathcal{X}} \subseteq \widetilde{\mathcal{X}}$ is the subset of memory deemed relevant to τ , returned in a form suitable for prompt augmentation or downstream computation.

By recalling only $\widehat{C}$, the agent conditions subsequent reasoning on task-specific context while keeping the effective prompt window compact and noise-free.

Continuing the shopping assistant example for cake ordering, suppose at turn $t+1$ the planner agent formulates the sub-goal $\tau = \text{“find gluten-free chocolate cake designs”}$. Invoking $\widehat{C} = Q(\Omega_t, \tau)$ returns a minimal set of memory entries, e.g.

$$\widehat{C} = \{\text{guest_allergy: gluten, child_pref: chocolate}\},$$

filtered from dozens of prior dialogue snippets. These facts are appended to the prompt fed into the CakeSearch tool, ensuring that the retrieval agent queries only gluten-free, chocolate-flavoured options while ignoring unrelated historical details. Thus, Q provides precise, context-aware recall that prevents irrelevant or outdated memories from polluting the reasoning chain.

For recommendation, the query τ supplied to Q is typically a ranking or explanation sub-goal, and the retrieved subset $\widehat{C}$ directly conditions the candidate set, the ranking, or the generated justification. Two recsys-specific failure modes follow. First, retrieving outdated or irrelevant traces can inject spurious constraints into ranking, for example resurfacing a preference the user has since abandoned. Second, retrieving privacy-sensitive history that should have been excluded can produce recommendations or explanations that are correct yet non-compliant. Effective retrieval in a recommender is therefore judged not only by topical relevance, as in the cake example, but by whether the recalled context preserves active constraints, excludes stale or deleted items, and respects the policy set π .

By introducing these formalisms, we establish a common language for discussing agentic behavior and capabilities. In the subsequent sections, we will demonstrate how these formal concepts can be instantiated and extended to create advanced recommendation systems and other multi-step, context-aware applications.

2.4 What Is Not an LLM Agent

Not every LLM-based recommendation pipeline is agentic. A *simple Q&A* or *single-prompt recommendation* system, such as an LLM prompted with “recommend five movies for this user”, is a generator or ranker, not an agent: it produces one response without persistent task state, autonomous tool choice, or iterative goal tracking. Similarly, a one-pass *retrieval-augmented generation* (RAG) system is a grounded generator, but not necessarily an agent, because the retrieve-then-generate sequence is fixed rather than selected, repeated, or revised by the model. For example, a RAG recommender may retrieve product descriptions and generate an explanation, but it does not decide on its own to re-query the catalog, ask a clarifying question, or invoke a policy checker.

The litmus test for *full* autonomy is whether the system maintains task state and can execute an observe–plan–act–verify loop in which it chooses its own actions. In recommender systems, this means that the system can decide whether to retrieve additional item evidence, ask a clarifying question, update or query memory, revise a ranking, invoke a policy or constraint checker, or stop because the recommendation objective has been satisfied. By this test, simple Q&A, one-pass RAG, fixed prompt chains, and static LLM-as-ranker pipelines are not autonomous agents, because their steps, branching rules, stopping criteria, and tool calls are predetermined rather than selected at run time.

Autonomy, however, is only one of the four capabilities of Section 2.2, and it is useful to separate it from the others. Planning and task decomposition, specialization across roles, tool use, and memory can each be present in a system whose control graph is fixed in advance. Consider the birthday-planning example of Figure 1. One could implement it as a fixed pipeline that always retrieves cake candidates, then decoration candidates, then party favors, checks the assembled set for theme and dietary consistency, and finally ranks it, with this order wired in at design time. Such a pipeline still decomposes the goal into specialized subtasks, calls retrieval tools, and carries the gluten-free constraint forward as a piece of memory, yet it never decides on its own to reorder these steps, ask the user a new question, or stop early. We therefore treat “agentic” as graded rather than binary: this fixed-graph version is agentic in *structure*, through decomposition, specialization, tool use, and memory, while remaining non-autonomous in *control*. It becomes fully agentic only when embedded in a closed-loop controller that can maintain state, choose its next action, inspect intermediate outcomes, and adapt the process toward the goal, for example by deciding to elicit a missing dietary restriction before retrieval, or to re-run decoration retrieval after the user enlarges the cake.

This structure-versus-control distinction is not merely terminological. The two dimensions can be varied independently, and the controlled study of Section 6 exploits this by holding autonomy fixed and varying only structure, so that the contribution of decomposition, specialization, and aggregation to recommendation quality can be measured on its own.

In short, not all LLM-based pipelines are agentic, and among those that are, not all exercise the same capabilities. Simple Q&A, one-pass RAG, fixed prompt chains, and static LLM-as-ranker pipelines can be useful components of an agentic recommender but are not sufficient on their own.

2.5 Scope: what is recommendation-specific in an agentic system?

An agentic recommender system contains both recommendation-specific and auxiliary components. We treat the following as core recommendation operations: candidate generation, evidence retrieval over user/item data, filtering and constraint satisfaction over item attributes, ranking and re-ranking, bundle or slate construction, explanation grounded in item and user evidence, feedback interpretation, memory update for personalization, and evaluation of recommendation utility. We treat natural-language parsing, generic task planning, tool routing, and user-interface generation

as adjacent capabilities: they may be necessary to implement an agentic recommender, but they are not themselves sufficient to constitute a recommendation contribution. This boundary matters because a multi-agent architecture can be impressive as an orchestration system while still failing at recommendation objectives such as relevance, diversity, calibrated personalization, fairness of exposure, or long-term user value.

Accordingly, throughout the paper we distinguish three layers: (i) the *recommendation layer*, responsible for user-item matching, ranking, and slate construction; (ii) the *interaction layer*, which manages dialogue, clarification, explanation, and preference elicitation; and (iii) the *orchestration layer*, which chooses tools, coordinates agents, manages memory, and verifies intermediate states. A system can be sophisticated at the interaction and orchestration layers yet add little at the recommendation layer, which is precisely the case the boundary above is meant to expose.

2.6 Positioning Relative to Non-Agentic and Agentic Recommenders

Table 2. Positioning of this paper relative to non-LLM, LLM-based, single-agent, and multi-agent recommender systems. ● = explicit/central; △ = partial or emerging; ○ = not central. Column meanings and the gap notes G1–G10 are given in the text.

Representative line / examples	Agent role			Agentic capability				Evaluation		Gap
	Assist.	Recomm.	Simul.	Memory	Tools / RAG	Planning	Coord.	Ranking	Trace	
Part I: Non-LLM recommender systems										
Classical / sequential recommender systems [29, 32, 33, 63]	○	●	○	△	○	○	○	●	○	G1
Conversational / interactive recommender systems [11, 23]	△	●	○	△	△	△	○	●	△	G2
Part II: LLM-based recommender systems, not explicitly agentic										
LLM as encoder, ranker, reranker, or reasoning module [79, 93]	○	●	○	△	△	○	○	●	△	G3
Generative / prompt-based recommender systems [13, 39]	○	●	○	△		△	○	●	△	G4
Part III: LLM-based recommender systems, single-agent										
Agent-assisted recommender [31, 92]	●	△	○	●	●	●	○	●	△	G5
Agent-as-recommender [76]	△	●	○	●	●	●	○	●	△	G6
Agent-as-user-simulator [9, 75, 89]	○	○	●	●	△	●	○	△	●	G7
Part IV: LLM-based recommender systems, multi-agent										
Multi-agent agent-assisted recommender [17, 77]	●	△	○	●	●	●	●	●	●	G8
Multi-agent agent-as-recommender [77, 90]	△	●	○	●	●	●	●	●	△	G9
Multi-agent user / environment simulation [5, 75]	○	○	●	●	△	●	●	△	●	G10
Part V: This paper										
Unified RecSys-specific framework and evaluation agenda	●	●	●	●	●	●	●	●	●	–

Sections 2.1 through 2.5 defined an agentic recommender system, separated it from LLM pipelines that are not agentic (Section 2.4), and bounded it against orchestration that is not recommendation-specific (Section 2.5). We close the section by placing this definition on the map of existing work. Table 2 organizes representative lines of research along three axes. The *agent role* axis records whether the agent assists an existing recommender (Assist.), acts as the recommender (Recomm.), or simulates users and environments (Simul.). The *agentic capability* axis records explicit memory (working/episodic/semantic/procedural), tool or retrieval use (Tools/RAG), planning or plan-act-verify behavior, and inter-agent coordination (Coord.). The *evaluation* axis records whether a line is assessed only on final ranking quality (HR@K, NDCG@K, MRR, diversity, exposure) or also on the intermediate agent trace (evidence grounding, memory use, tool correctness, judge reliability,

communication cost, latency, and failure attribution). These axes separate three issues that are often conflated: that a system may use an LLM without being agentic, that an agentic recommender may differ in role and in which capabilities it exercises, and that evaluation may stop at the ranked list or extend to the trace. The table therefore makes precise why this paper is neither a survey of LLM-based recommendation nor a generic multi-agent-system discussion, but a RecSys-specific framework for specifying and evaluating agentic recommendation architectures.

The capability columns correspond directly to the primitives of the formal framework: Memory is the store Ω with its operators $\mathcal{U}$ and $\mathcal{Q}$ (Definitions 2.3 and 2.4), Tools/RAG is the function set $\mathcal{F}$ of Definition 2.1, Planning is the decomposition behavior of Section 2.2, and Coordination is the protocol Π of the MAS triple (Definition 2.2). The rows then trace a progression, and the Gap column names what each stage still leaves open. Classical and sequential recommenders (G1) provide strong ranking foundations but expose no agent loop, and conversational recommenders (G2) are interactive yet usually scripted rather than tool-using and memory-bearing. LLM-based but non-agentic recommenders improve representation, ranking, or generation while remaining inside fixed pipelines, so autonomy and tool choice stay implicit (G3, G4). Single-agent systems introduce memory, tools, and planning, but agent-assisted designs lack inter-agent specialization (G5), agent-as-recommender designs often leave the memory, tool, and planning contributions under-ablated (G6), and agent-as-simulator designs face fidelity, calibration, and label-leakage difficulties (G7). Multi-agent systems add coordination through Π , yet the marginal utility and cost of extra agents is rarely attributed (G8), decomposition and verification raise communication complexity (G9), and richer simulation can amplify artificial behavior or simulator drift (G10). The final row marks the gap this paper targets: a unified, recommender-specific vocabulary and evaluation agenda in which agentic complexity is justified through measurable gains in recommendation quality, grounding, trace reliability, fairness, cost, and robustness rather than assumed, with the under-ablation gaps G6 and G8 addressed empirically in Section 6.

3 Memory Storage and Retrieval Mechanisms

Due to importance of memory mechanisms in agentic systems, especially those anchored toward recommender system tasks we will discuss these mechanisms in a dedicated section. Large Language Model (LLM) agents require robust memory mechanisms to behave coherently over time. By default, an LLM is stateless, which means that each query is processed independently, with no built-in recall of previous interactions. This stateless design is a serious issue in hindering LLMs to be adapted to recommendation tasks requiring personalization and “remembering” things about users.

To overcome this, agent frameworks introduce explicit memory systems that let the agent “remember” and reuse information from past events or dialogues. Broadly, these memories fall into short-term (working memory) and long-term categories, with further distinctions (episodic, semantic, procedural, etc.) inspired by human cognition. This section surveys the types of memory in LLM agents and the storage/retrieval mechanisms.

3.1 Types of Memory in LLM Agents

3.1.1 Short-Term (Working) Memory. Short-term (often called “working” [24]) memory refers to the transient context the agent holds in mind. In an LLM agent, this corresponds to the recent interaction history or “context window” provided to the model on each turn. It is analogous to human working memory in that it is readily accessible but of limited capacity (bounded by token length) [65]. For instance, a chatbot’s short-term memory might include the last few user prompts and agent responses, allowing for conversational coherence. Short-term memory is therefore crucial for immediate reasoning, but it does not persist beyond the current session (i.e., it lacks a true long-term store).

3.1.2 Long-Term Memory. Long-term memory provides persistent recall across interactions and over time. In agent systems, it can be subdivided into:

Episodic Memory Episodic memory pertains to specific events or experiences (*episodes*) the agent has encountered. In LLM agents, this often means a stored log of past dialogues, observations, or actions, coupled with context such as timestamps or metadata. It is context-rich and instance-specific. In this case, the agent remembers not just *what* happened, but *when* and *under what circumstances* [22]. For instance, the agent might recall: *Last week the user asked about Italian restaurants and I recommended XYZ*. Episodic memory supports one-shot learning of events and explicit recall of past episodes, and some researchers argue it is a key missing piece for truly long-lived LLM-based agents. [59]

Semantic Memory Semantic memory stores general facts, concepts, or knowledge about the world, akin to the “know-what.” In LLM agents, semantic memory might be an external knowledge base or database of facts (e.g., “in my retrieval system TVs are categorized under electronics category”) [88].

While much semantic knowledge is encoded in the LLM’s pretrained weights, agents can also maintain explicit semantic stores (relational databases, knowledge graphs, etc.) for dynamic updates. Semantic memory is *long-term* and *explicit*, though it abstracts away the specific instance in which a fact was learned [35]. For example, from several chats the agent might distill a high-level fact: “This user prefers Italian cuisine.”

Procedural Memory Procedural memory, or the “know-how”, refers to skills or procedures the agent has learned to execute without needing to deliberate each time. For an LLM agent, this might appear as learned prompts, scripts, or code snippets that automate common tasks (e.g., connecting to a database). [71, 78]

Procedural memory is typically implicit in that it is not stated as factual knowledge but manifested as rules or sequences of actions. It allows the agent to execute tasks more efficiently, refining these skills through repeated practice. Early cognitive architectures like SOAR [34] used production rules to represent procedural knowledge, and modern LLM agents similarly encode procedural behaviors through fine-tuned prompts, scripted tool invocations, or learned action sequences that mimic rule-based execution pipelines[12, 53]. In the next subsection, we explore how these different types of memory are realized in the actual agentic recsys implementations.

3.2 Memory Storage and Retrieval Mechanisms

Large-context LLMs can “remember” only a few thousand tokens per forward pass, yet real recommender deployments routinely span months of user interaction and terabytes of product data. Consequently, an *agentic* recommender must decide “what” to store externally, “how” to distill it, and “which” fragments to re-inject into the prompt at inference time. We formalise these operations through the **Memory Update** and **Memory Retrieval** functions (Definitions 2.3–2.4), then survey concrete storage modalities. For having a more complete survey of the memory mechanisms, first we state the following definitions.

Definition 3.1 (Memory Item). A memory item is a triple $m = (k, v, \varsigma)$ where

- (i) $k \in \mathbb{R}^{d_k}$ is the **key**, typically an n -gram embedding or hashed identifier that serves as the retrieval handle;
- (ii) $v \in \mathbb{R}^{d_v}$ is the **value** payload that preserves the semantic content (natural-language text or a dense vector);
- (iii) $\varsigma = (t, \ell, u)$ is a metadata tuple storing the time-stamp t , a logical label $\ell \in \{\text{EPI}, \text{SEM}, \text{PROC}\}$ distinguishing *episodic*, *semantic*, or *procedural* memory, and an update counter u that tracks how often the item has been modified.

For instance, in our cake retrieval example, the statement of the user for gluten allergy, “guest_allergy = gluten” is stored as

$$k = \text{embedding}(\text{“gluten allergy”}), \quad v = \text{“gluten”}, \quad \varsigma = (t=17:45, \ell = \text{EPI}, u = 1).$$

Here k is an embedding representation of the key “gluten allergy”; $\ell = \text{EPI}$ signals that the fact is tied to a specific party event rather than a timeless preference.

The key–value–metadata form in Definition 3.1 is an analytic wrapper, not a restriction on implementation. The value field may contain raw text, dense embeddings, symbolic triples, relational rows, procedural scripts, multimodal features, pointers to external stores, or other memory representations. The purpose of the wrapper is to expose the three pieces of information that agentic recommendation needs from any stored item: a retrieval handle, a semantic payload, and metadata for recency, provenance, privacy, and update control. In a recommender these metadata fields are not incidental bookkeeping. The timestamp t determines whether a preference is still current or stale, the label ℓ separates a one-off event such as a party constraint from a durable taste, and the update counter u records how often a fact has been reinforced, which is exactly the signal needed to decide whether a preference is worth conditioning a ranking on.

Definition 3.2 (Relevance Scoring). Given a task query $\tau \in \mathcal{T}$ and a memory item $m = (k, v, \varsigma)$, a *relevance function*

$$S : \mathcal{T} \times (\mathcal{X} \times \tilde{\mathcal{X}} \times \mathbb{R}^3) \longrightarrow \mathbb{R}_{\geq 0},$$

returns a non-negative score. Retrieval (Definition 2.4) selects the K highest-scoring items:

$$\hat{\mathcal{C}} = \text{Top-}K_{m \in \Omega} S(\tau, m).$$

Again, back to our cake retrieval example, let $\tau = \text{“find gluten-free chocolate cake”}$. Then, we can choose the relevance function in a form like $S(\tau, m) = \cos(\text{embedding}(\tau), k)$. The cosine term elevates items semantically close to the query. Consequently, the system surfaces the allergy item and the child’s flavor preference, but not unrelated recommendation traces.

In a recommender, topical similarity alone is an impoverished relevance function. What should be recalled for a ranking or explanation sub-goal depends not only on whether a memory item is semantically close to τ , but on whether it is still valid, whether it encodes a binding constraint, and whether it is permitted to be used. A useful S for agentic recommendation therefore combines the similarity term above with the metadata of Definition 3.1: a recency weight derived from t so that abandoned preferences decay, a label term from ℓ so that hard constraints (a dietary restriction, a returned item) are retrieved with priority over soft tastes, and a privacy gate that excludes items the user has asked to delete. This is what distinguishes memory relevance in a recommender from generic semantic search: the highest cosine score is not always the item that should condition the next ranking.

Now, we will review different modalities that memory store/retrieve can take.

3.2.1 Raw Text Logs. The simplest store is an *append-only* transcript $\Omega^{\text{raw}} = [m_1, \dots, m_T]$, where each m_t is the verbatim turn text. Retrieval is concatenation of the last L tokens: $\hat{\mathcal{C}} = \text{Tail}_L(\Omega^{\text{raw}})$. Although the update function (Definition 2.3) $\mathcal{U}$ is trivial (append), this scales poorly and prompt length soon exceeds the model window. [3, 43]

3.2.2 Summarisation (Compression). In this approach, older logs are periodically compressed into shorter summaries. More specifically, update function $\mathcal{U}$ invokes an LLM summariser, $\mathcal{R}_{\text{sum}} : \mathcal{C}_t \rightarrow \tilde{\mathcal{C}}_t$, which compresses $\mathcal{C}_t$ to a more concise context $\tilde{\mathcal{C}}_t$. The agent then stores and retrieves these summaries to preserve essential information. Though it saves tokens, the fidelity of recall

depends on the quality of summarization. Recent memory compression methods enable adaptive summarization and storage of long conversational histories for LLM agents [40].

3.2.3 Embedding and Vector Databases. For long-term memory, the prevailing method in retrieval-augmented generation (RAG) is embedding-based storage. Each memory item is encoded as a high-dimensional vector (using an embedding model e) $k = e(v)$, $v = (\text{raw text})$ and stored in a vector database [21, 86]. For Retrieval, using the retrieval function Q (see Definition 2.4) the current query is embedded, and nearest-neighbor search surfaces semantically relevant items, enabling large-scale, flexible recall—though typically at the cost of strict chronological ordering. Please note that classic retrieval-augmented models RETRO [6] and REALM [25] demonstrate the scalability and effectiveness of such large-scale external memory designs.

The choice among these modalities is a recommendation-deployment decision, not only an engineering one. Vector stores scale to catalog- and user-history sizes and tolerate fuzzy recall, which suits personalization, but they blur chronology, which matters when the order of a user’s interactions carries intent. Raw logs and summaries preserve session dynamics but do not scale to months of history. Structured stores enforce the hard constraints and provenance that policy-compliant recommendation requires. In practice an agentic recommender mixes them by role: short-term session state in a buffer, durable tastes in a vector store, and constraints and provider or policy facts in a structured store where they can be checked exactly.

3.2.4 Knowledge Graphs / Structured Stores. Beyond raw text or vector embeddings, an agent may persist “knowledge–graph memory” in the form of symbolic triples (s, r, o) (subject–relation–object) or relational table rows that can be queried explicitly via SQL or SPARQL. Symbolic storage is particularly well-suited for semantic memory: it supports precise retrieval, logical entailment, and integrity constraints, albeit at the cost of additional curation.

Definition 3.3 (Symbolic Memory Item). A symbolic memory item augments Definition 3.1 by taking $m = (k, v, \zeta)$ with $v = (s, r, o) \in \mathcal{E}^3$, where $\mathcal{E}$ is the entity set of the knowledge graph. In other words, the value can be of the form subject–relation–object for a given query. Equivalently, one can assume an embedding on the triplet for key retrieval used for approximate nearest-neighbour fallback.

For example, suppose the assistant stores $v = (\text{user}, \text{likes}, \text{chocolate})$ with label $\ell = \text{SEM}$. Given user query $\tau = \text{“What flavour cake should I order?”}$ the system formulates $(\text{user}, \text{likes}, x)$ and retrieves the triple, yielding the concrete value chocolate. If no exact match exists, the agent can still fall back to the embedding key $k = \text{embedding}(\text{user}, \text{likes}, \cdot)$ and perform a vector search for semantically close preferences.

Symbolic knowledge graph/structured stores offer high-precision constraints and explainable provenance [45], but demand manual or automated pipelines to populate and maintain the triples; they also introduce schema-evolution overhead when the domain ontology changes. [85]

3.2.5 Parametric Memory Updates. A costly alternative is to integrate $\tilde{\mathcal{C}}_t$ directly into the backbone parameters of the LLM model [19]. Formally, the update function $\mathcal{U}$ returns a new language model $\mathcal{M}_{t+1} = \mathcal{M}_t \oplus \Delta_t$, where $\mathcal{M}_t$ represents the model at time t and Δ_t represents the update on the model to incorporate the new events. This makes Ω implicit in $\mathcal{M}$. Latency and safety constraints limit real-time use [10, 20].

Regulated Context Windows. In production, agents often blend *procedural*, *episodic*, and *semantic* memories with a regulator:

$$\widehat{\mathcal{C}} = \widehat{\mathcal{C}}_{\text{PROC}} \cup \widehat{\mathcal{C}}_{\text{SEM}} \cup \widehat{\mathcal{C}}_{\text{EPI}},$$

Table 3. Use cases for agentic recommender systems. The table summarizes the four representative scenarios developed in Section 4, the practical trigger for using an agentic design, the recommendation-layer gain, and when a simpler non-agentic baseline may be sufficient.

Use case	Agentic trigger	Recommendation-layer gain	Simpler baseline may suffice when
Interactive goal-oriented recommendation	Clarification, planning, memory, and bundle verification	Improves constraint satisfaction, bundle coherence, ranking quality, personalization, and user effort in multi-turn recommendation.	The request is a single well-scoped query with a stable user profile and fixed candidate set.
User simulation and evaluation	Synthetic users, logging, evaluators, and session summarization	Supports offline stress testing, counterfactual diagnosis, robustness checks, and evaluation before live deployment.	Reliable logged feedback is already available and no interactive trajectory or simulated behavior needs to be modeled.
Contextual and multi-modal recommendation	Vision tools, external retrieval, memory, and consistency checks	Connects images, context, metadata, user preferences, and spatial or aesthetic constraints to ranking and bundle construction.	All relevant features are already structured and can be consumed directly by a conventional ranker.
Grounded recommendation explanation	Evidence retrieval, critic/evaluator agents, and policy checks	Produces explanations that are more faithful, evidence-backed, policy-consistent, and less prone to unsupported claims.	A short template explanation is sufficient and dynamic evidence or policy checking is unnecessary.

subject to a token budget $|\widehat{C}| \leq B$ [81]. We can formalize this type of regulator as a knapsack problem (see [51]) variant

$$\begin{aligned} \max_{\widehat{C} \subseteq \Omega} \quad & \sum_{m \in \widehat{C}} S(\tau, m) \\ \text{s.t.} \quad & \sum_m |m| \leq B. \end{aligned}$$

where $S(\tau, m)$ is the relevance score as defined in definition 3.2, $|m|$ is the length of the retrieved memory, and B is the token budget.

The foregoing definitions and examples demonstrate that memory in agentic recommender systems is no longer a monolithic cache but a spectrum of interlocking mechanisms—from raw transcript buffers and lossy summaries to vector stores to fully structured knowledge graphs, to budgeted tokens. By casting memory operations as explicit update $\mathcal{U}$ and retrieval $\mathcal{Q}$ functions, parameterised by relevance scores and symbolic query semantics, we obtain a principled foundation for analyzing capacity, latency, and factual fidelity. Yet the formalism also exposes substantial gaps: adaptive compression that preserves downstream utility is still heuristic; relevance scoring lacks theoretical guarantees; and symbolic stores incur unsolved curation and schema-evolution costs. Bridging these gaps will require new learning objectives that couple retention with task performance, tighter integration of uncertainty calibration into retrieval, and hybrid architectures that combine the precision of knowledge graphs with the scalability of dense embeddings. For recommendation specifically, these gaps surface as measurable failure modes: compression that drops a binding constraint, relevance scoring that resurfaces a stale preference, and retrieval that returns a deleted trace. The memory-correctness, staleness, and deletion-compliance signals of Section 5 are intended to quantify exactly these failures. In short, memory for LLM-driven, multi-agent RecSys is both a critical enabler and an open frontier, inviting further research at the intersection of information retrieval, knowledge representation, and large-scale language modeling.

4 Agentic Recommendation Task Families

The remainder of this section grounds the general agentic framework in four representative use cases, each chosen to illuminate a distinct facet of next-generation recommender systems. As summarized in Table 3, these examples are not intended to be exhaustive; rather, they show when planning, memory, tool use, verification, or multi-agent coordination can improve recommendation-layer outcomes, and when a simpler non-agentic baseline may be sufficient.

4.1 Interactive Recommendation

Formal Task Definition. We define *Interactive Recommendation* as the problem setting in which an agent (or a set of collaborating agents) engages in a multi-turn dialogue with a user to identify, refine, and present suitable recommendations. Unlike static recommender systems that rely on one-shot user inputs, interactive recommendation leverages iterative exchanges to incorporate user feedback, contextual constraints, and additional information drawn from memory or external resources. The overarching objective is to dynamically adapt suggestions as the conversation evolves, thereby providing a more personalized and contextually appropriate set of recommendations.

Let $\mathcal{D}$ be the space of dialogue turns and $\mathcal{S}$ the universe of documents (in an eCommerce setting these can be purchasable stock-keeping units, SKUs). A user session at step t is characterised by the partial transcript

$$C_{1:t} = (d_1, a_1, \dots, d_{t-1}, a_{t-1}, d_t),$$

where $d_i \in \mathcal{D}$ is a user utterance and $a_i \in \mathcal{D}$ the agent reply. Define the *interactive recommendation task* as a mapping

$$\Phi : (C_{1:t}, \mathcal{E}) \longrightarrow \langle s_{(1)}, \dots, s_{(L)} \rangle,$$

that outputs a ranked list of L items with the maximal expected utility $\sum_{j=1}^L \text{Rel}(s_{(j)} \mid C_{1:t})$, conditioned on user constraints (theme, dietary rules, budget) implicitly encoded in $C_{1:t}$.

To illustrate this task, we consider the scenario of planning a Mickey Mouse-themed birthday party as shown in Figure 1. A parent consults the system to select decorations, arrange a suitable cake, and accommodate guests' dietary restrictions. The agent aims to guide the parent through each step, from confirming the child's preferences (such as color schemes or favorite cake flavors) to identifying any special requirements. By interacting iteratively with the user and retrieving relevant information from its memory and external tools, the agent can refine its suggestions over time.

High-level goal. The agent must *adaptively refine* Φ through multi-turn dialogue, spawning specialized sub-agents and tool calls to satisfy latent sub-goals (e.g. "select gluten-free chocolate cake", "choose decor consistent with Mickey-Mouse palette") while preserving conversational coherence and minimizing user effort.

System architecture and agent setup. For this specific example, we can instantiate a multi-agent system

$$\text{MAS}_{\text{party}} = (\mathcal{A}, \mathcal{E}, \Pi),$$

where **Agent set** $\mathcal{A} = \{A_{\text{chat}}, A_{\text{epi}}, A_{\text{nli}}, A_{\text{SAC}}, A_{\text{cake}}, A_{\text{decor}}, A_{\text{favor}}, A_{\text{col_check}}, A_{\text{rank}}\}$. Each A_i is an LLM agent in the sense of Definition 2.1. Where

- A_{chat} (chat agent) primary interface to the user.
- A_{epi} – episodic retrieval using Q (Definition 2.4).
- A_{nli} – natural-language inference to vet relevance of retrieved episodes.
- A_{SAC} – specialised-agent caller that spawns three micro-MAS blocs: $\{A_{\text{cake}}, A_{\text{decor}}, A_{\text{favor}}\}$ for category-specific retrieval.

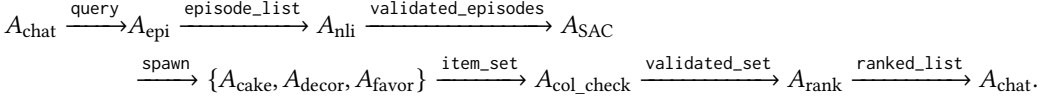
- $A_{\text{col_check}}$ – collection-level consistency.
- A_{rank} – personalised ranking & presentation.

Also one can define the Environment for this MAS as:

$$\mathcal{E} = (\text{ProductCatalogue}, \text{UserProfileDB}, \text{VectorDB}, \text{LayoutTool}),$$

and communication protocol as $\Pi = (\mathcal{C}, \Gamma)$ where $C_{ij} = 1$ iff either A_j is a child spawned by A_i or $A_i = A_{\text{chat}}$. Γ contains message types {query, episode_list, tool_call, item_set, ranked_list}.

The following also illustrates communication sketch:



Note that spawn can be autonomous depending on $\mathcal{C}$

Memory and tool requirements. Now, we illustrate how each type of memory is defined here and how each agent can utilize it. Short-term (Working) memory can be defined for A_{chat} that takes logs of last L turns of the $C_{1:t}$. Episode store can be accessed and manipulated by A_{epi} through $\mathcal{U}$ and $\mathcal{Q}$. Semantic memory Ω^{SEM} can track stable user traits (preferred colours, dietary restrictions). Ω^{PROC} can include repeatable prompt templates for each retrieval block as well as information about each table in DB so that agents can autonomously query them or other elements of environment $\mathcal{E}$. External Tools $\mathcal{F}$ in this MAS are SearchCakeAPI, SearchDecorAPI, SearchFavorAPI, VectorDB.query (semantic retrieval), LayoutTool.generate (graphic board of decor set).

Benefits and discussion. The architecture realises *interactive recommendation* by decomposing a fuzzy, high-level goal (“celebrate my child’s Mickey-Mouse birthday”) into tractable sub-tasks handled by specialised agents. Key advantages like (i) Modularity, where each retrieval block is itself a micro-MAS that can be re-used for other party themes or plugged into A/B tests without retraining the entire pipeline. (ii) Memory-aware personalisation, where episodic retrieval (A_{epi}) injects user- and session-specific constraints (gluten-free, colour palette) while semantic memory supplies timeless preferences, yielding recommendations that are both relevant and surprising. (iii) Error containment, where the NLI validator and collection-consistency agent act as *fact gates*, reducing hallucination propagation by rejecting items that violate theme constraints or dietary rules, (iv) autonomy, where the specialised-agent caller spawns retrieval workers on demand and per query, and (v) Rich user experience where the final ranked set, together with a layout generated by LayoutTool, offers a cohesive shopping board, demonstrating how agentic orchestration can move beyond single-item suggestions to holistic, story-driven recommendations.

Collectively, the design illustrates how the formal primitives—MAS, $\mathcal{U}$, $\mathcal{Q}$, and typed memory stores—translate into a concrete, deployable pipeline for next-generation conversational shopping assistants, which far exceeds the capabilities of classic recommender systems.

4.2 User Simulation and Recommendation Evaluation

Formal definition of the task. User simulation and recommendation evaluation together form a framework in which synthetic user behavior is generated to probe, test, and refine recommender systems before full-scale deployment. Let $R_\phi : \mathcal{X} \rightarrow \mathcal{S}^L$ denote a recommender parameterised by ϕ that maps the current session state $x \in \mathcal{X}$ to an ordered list of L items from the catalogue $\mathcal{S}$. A **user simulator** is a stochastic policy

$$\mathcal{M}_{\theta, \omega} : \mathcal{X} \times \mathcal{S}^L \longrightarrow \mathcal{A}_{\text{user}}, \quad \mathcal{A}_{\text{user}} \in \{\text{Select}, \text{Not Select}\},$$

parameterised by intrinsic preference vector θ and noise vector ω . Note that user simulator's action space $\mathcal{A}_{\text{user}}$ can include another set of options depending on the recommendation setting. For instance it can be {Click, Pass, Purchase} in eCommerce user simulation. Given repeated interaction over T time steps, the joint process $\{x_t, s_t^{(L)}, a_t^{\text{user}}\}_{t=1}^T$ induces an empirical evaluation functional

$$\Psi(R_\phi, \mathcal{M}_{\theta, \omega}) = \mathbb{E} \left[\sum_{t=1}^T g(x_t, s_t^{(L)}, a_t^{\text{user}}) \right],$$

where $g(\cdot)$ is a reward such as Select or diversity penalty. The “simulation-based evaluation problem” seeks to estimate Ψ under controlled distributions of θ and ω .

High-level goal. In practical scenarios, gathering data from real users can be expensive or infeasible, especially when conducting A/B tests for newly introduced features. A simulated user that approximates realistic browsing, clicking, and purchasing behaviors offers a controlled environment for stress-testing various recommendation strategies. For example, an e-commerce platform aiming to recommend Bohemian-style furniture may wish to simulate how a typical user navigates through the site, examines multiple items, compares prices, and ultimately decides whether to purchase. This allows the platform to estimate click-through rates and conversions without requiring a large real-user pilot study. The primary aim is to replicate the interactions of a diverse user population in order to evaluate and compare recommendation strategies. The system must produce synthetic but plausible session data, track user decisions at each step, and generate summary statistics or qualitative insights about overall performance. By testing both short sessions (quick browsing) and long sessions (prolonged decision-making), one can obtain a more comprehensive understanding of the strengths and weaknesses of the underlying recommender algorithms.

System architecture and agent setup. We instantiate a multi-agent system (also see Figure 2)

$$\text{MAS}_{\text{sim}} = (\mathcal{A}, \mathcal{E}, \Pi), \quad \mathcal{A} = \{A_{\text{rec}}, A_{\text{user}}, A_{\text{note}}, A_{\text{eval}}, A_{\text{summ}}, A_{\text{report}}\}.$$

- A_{rec} (*Recommender Agent*) is an LLM agent that wraps R_ϕ and orchestrates tool calls (SearchAPI, SessionTracker).
- A_{user} (*User Simulator*) implements $\mathcal{M}_{\theta, \omega}$; it possesses a structured memory Ω^{SEM} of preferences and Ω^{STM} for the current session trajectory.
- A_{eval} (*Evaluation Agent*) logs tuples $(x_t, s_t^{(L)}, a_t^{\text{user}})$ via update function $\mathcal{U}$ -style appends to long-term storage. It computes instantaneous metrics g_t by querying logs with Q and emits eval_event messages.
- A_{summ} (*Session Summariser*) invokes a compression operator $\mathcal{R}_{\text{sum}}$ every T turns to produce a summary vector of the session's salient outcomes.
- A_{report} (*Reporter*) aggregates thousands of summaries, performs statistical tests, and outputs a final PDF/CSV dashboard.

Environment $\mathcal{E} = (\text{ProductDB}, \text{PreferenceSampler}, \text{LogStore})$, provides catalogue metadata, draws synthetic θ , and persists interaction traces.

Communication Protocol matrix of this MAS, C allows the sequence

$$A_{\text{rec}} \rightleftarrows A_{\text{user}} \rightarrow A_{\text{eval}} \rightarrow A_{\text{summ}} \rightarrow A_{\text{report}},$$

with message types $\Gamma = \{\text{rec_list}, \text{user_action}, \text{log_entry}, \text{eval_event}, \text{session_summary}, \text{final_report}\}$.

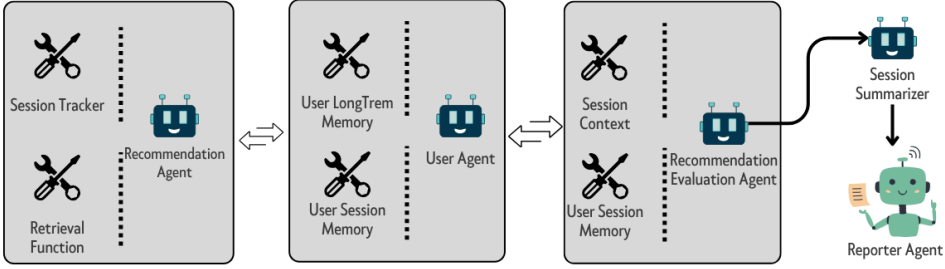


Fig. 2. A sample multi-agent system for user behavior simulation and recommendation evaluation

Memory and tool requirements. A_{user} can access and update semantic memory, Ω^{SEM} which includes latent taste vector, price sensitivity, Ω^{PROC} which stores navigation policy, click heuristics, and $(\Omega^{\text{EPI}}$ which remembers prior simulated sessions (for repeat-exposure effects). A_{eval} uses a raw text log plus vector store to support fast Q filters by item ID or action type. It also maintains sliding-window statistics (e.g. running CTR, diversity entropy) in Ω^{STM} . Tool access by other agents is illustrated in Figure 2.

Benefits and discussion. The stated MAS architecture can yield the following traits: (i) Cost-effective experimentation, where hundreds of parameterized user agents can be spawned in parallel, yielding reliable offline estimates of $\Psi(R_\phi, \mathcal{M})$ before any live-traffic exposure. (ii) Cold-start mitigation, where by sampling user profiles with sparse or unseen preference vectors, the framework stresses R_ϕ under low-data regimes and surfaces failure modes early. (iii) Diagnosis of systemic bias, where the reporter’s aggregate view highlights trends such as over-pricing, popularity bias, or demographic skew, which can be traced back to fine-grained logs for root-cause analysis. (iv) Memory-driven realism, where the division of semantic, episodic, and procedural memories within A_{user} allows long-range preference carry-over (“I still want a bohemian sofa”) while modelling bounded session attention—all expressed via the formal $\mathcal{U}/Q$ operators. (v) Modular extensibility, where new evaluation criteria (fairness, robustness) can be implemented by adding auxiliary evaluator agents without retraining the core recommender, consistent with the MAS abstraction.

Overall, this architecture turns user simulation into a first-class, memory-aware MAS that provides quantitative and qualitative feedback loops—bridging the gap between offline metrics and live user studies for next-generation recommender systems.

4.3 Contextual and Multi-Modal Recommendation

Formal definition of the task. We define the task as the process of generating suggestions that incorporate diverse forms of input, such as text descriptions, user interaction histories, and images depicting the user’s physical or aesthetic environment. For instantiation, let $\mathbf{x} \in \mathcal{T}$ denote textual constraints (e.g. “Bohemian, earthy colours”), $\mathbf{v} \in \mathcal{V}$ a set of one or more images encoding spatial context, and $\mathbf{u} \in \mathcal{N}$ a latent user-profile vector drawn from long-term memory. We model contextual, multi-modal recommendation as a mapping

$$\mathcal{R}_\phi : (\mathbf{x}, \mathbf{v}, \mathbf{u}) \longrightarrow \hat{\mathbf{y}} = \langle s^1, s^2, s^3, \dots \rangle,$$

where each $s^{(\cdot)} \in \mathcal{S}$ is a SKU and the ranked tuple $\hat{\mathbf{y}}$ maximises joint relevance $\sum_i \text{Rel}(s_i \mid \mathbf{x}, \mathbf{v}, \mathbf{u})$ subject to *complementarity* and *aesthetic-coherence* constraints. The latter can be enforced by an

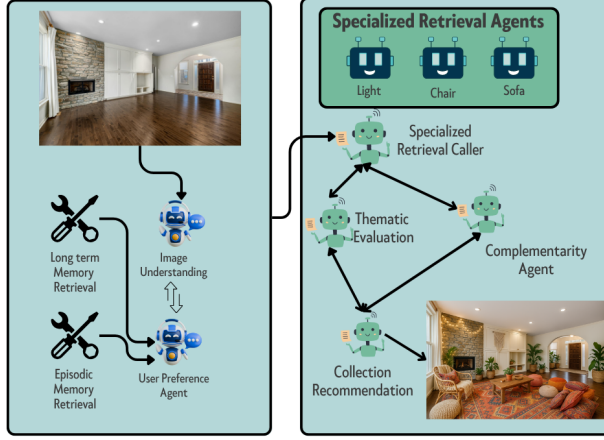


Fig. 3. An example of multi-agent pipeline for multi-modal recommendation.

auxiliary predicate $\text{Compat}(\hat{\mathbf{y}}) = 1$ if all selected items harmonise in palette and style, or any *Coherence Score* mechanism.

High-level goal. In many practical scenarios, recommendations cannot rely solely on textual inputs. A user interested in redecorating a living room may upload one or more photographs depicting the current layout, along with a brief description of desired styles (e.g., Bohemian, minimalist, rustic-see Figure 3). The system must analyze these images to identify available space, prevailing color palettes, and aesthetic elements, while also taking into account prior user behavior or known preferences (such as a history of purchasing similar items). By combining textual context with visual analysis, a multi-modal recommender can produce more cohesive suggestions, reduce guesswork, and streamline the user’s decision-making process. The ambition in an example orchestration is to replicate a professional interior-designer workflow while eliminating iterative, item-by-item searches.

System architecture and agent setup. The pipeline in Fig. 3 is formalised as $\text{MAS}_{\text{mm}} = (\mathcal{A}, \mathcal{E}, \Pi)$ with

$$\mathcal{A} = \{A_{\text{chat}}, A_{\text{image}}, A_{\text{history}}, A_{\text{cat}}, A_{\text{caller}}, A_{\text{ThemChk}}, A_{\text{compChk}}, A_{\text{collect}}\}.$$

A single conversation unfolds as follows. The chat agent A_{chat} receives $(\mathbf{x}, \mathbf{v})$ and forwards the image to A_{image} , which extracts palette vector $\mathbf{p}$ and spatial affordances via a vision backbone $\mathcal{F}_{\text{layout}} \in \mathcal{F}$. Concurrently, A_{history} retrieves $\hat{C}_{\text{SEM}} \subset \Omega^{\text{SEM}}$, stable style and budget preferences, using the retrieval operator Q . The specialised-retrieval caller A_{caller} fuses $\mathbf{x}, \mathbf{p}, \hat{C}_{\text{SEM}}$ and outputs a set of target classes like $C = \{\text{chair}, \text{sofa}, \text{lamp}\}$. For each $c \in C$, it spawns an intra-session micro-MAS. The resulting candidate lists are concatenated and passed to A_{ThemChk} , which rejects items violating user history (e.g. bans leather if the user is vegan). A_{compChk} then checks for complementarity score for the bundle of recommended items (maybe through solving mixed integer program). Finally, A_{collect} ranks the surviving set by personalised utility and calls $\mathcal{F}_{\text{render}}$ to generate an on-image mock-up before handing results back to A_{chat} .

The protocol’s communication matrix $\mathbf{C}$ is sparse. For instance, A_{chat} may message any agent, but checker agents communicate only downstream, preventing cyclic justification loops that might amplify hallucinations.

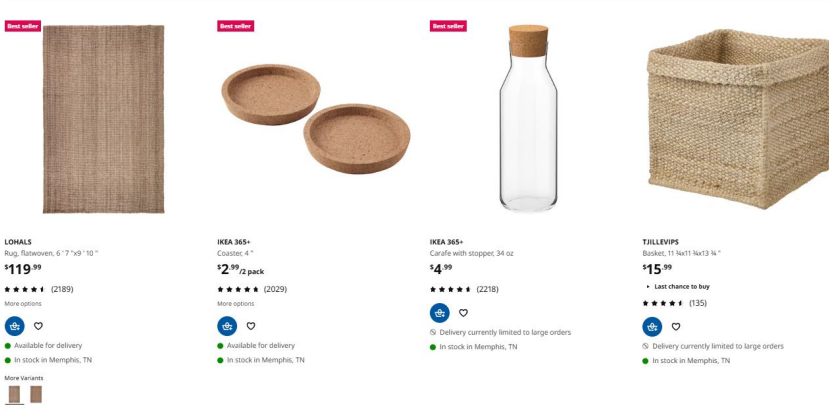


Fig. 4. A sample of coherent (Boho style with earthy colors) multi-category recommendation (courtesy ikea.com).

Memory and tool requirements. Short-term buffers hold the live prompt plus extracted palette vectors; Ω^{SEM} caches enduring preferences (colour, material, budget); Ω^{EPI} stores prior styling sessions so that the system can avoid repetitious suggestions. Key tools are the vision encoder, a vector-search API over ProductDB, and a layout renderer that embeds selected SKUs into the uploaded photograph.

Benefits and discussion. The multi-modal MAS realises a “one-shot designer” experience. Semantic and complementarity checks decouple *rule compliance* from *aesthetic harmony*, yielding recommendations that are both on-brand and visually pleasing. Finally, because the agent caller can spawn modular micro-MAS blocks live, the system stays flexible to a variety of user requests. In sum, contextual and multi-modal recommendation showcases how LLM-powered agents, external vision tools, and structured memories can collaborate to deliver designer-level curation in a single interactive session. Like the previous use case, this enables a capability beyond classic recommender systems.

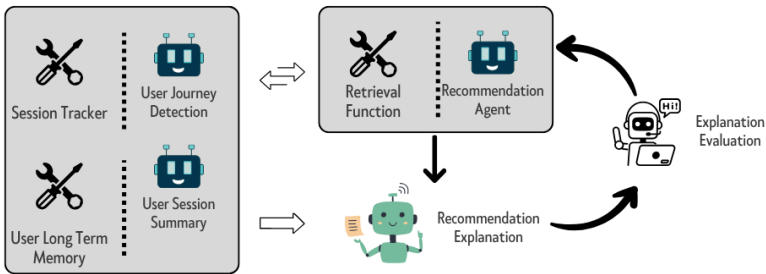


Fig. 5. Adaptive Multi-Agent Recommendation Explanation

4.4 Recommendation Explanation

Formal definition of the task. We define *Recommendation Explanation* as the process of generating intelligible and contextually relevant justifications for why particular items are recommended to a user (see [8]). Let $\mathbf{r} \in \mathcal{S}^L$ be a set of items surfaced by a recommender R_ϕ at dialogue step t and let $\mathbf{u}_t = (\widehat{C}_t^{\text{SEM}}, \widehat{C}_t^{\text{EPI}})$ denote the user’s current semantic and episodic context retrieved via Q . A *recommendation-explanation function* is a mapping

$$\Phi_\psi : (\mathbf{r}, \mathbf{u}_t) \longrightarrow e_t \in \mathcal{E}_{\text{text}}, \quad \text{s.t. } \text{Consistency}(e_t) = 1,$$

where ψ parameterises an LLM agent, e_t is a natural-language explanation, and Consistency is a predicate that checks (i) factual alignment with $\mathbf{u}_t$ and (ii) stylistic conformity to brand rules.

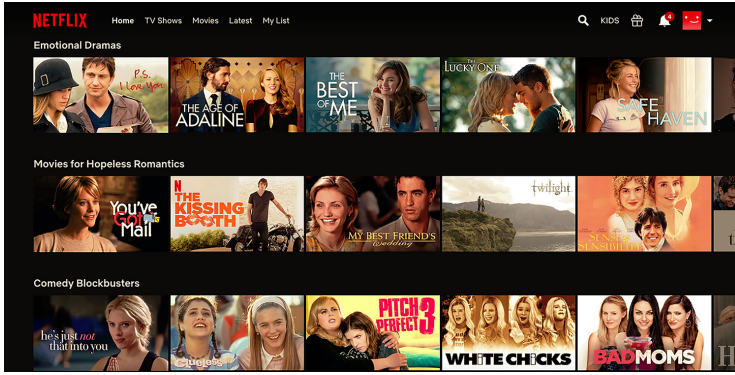


Fig. 6. Explaining Recommendations (e.g., Emotional Dramas in the first set of recommended movies) provides more transparency and helps with user engagements (courtesy netflix.com).

High-level goal. The system seeks to frame each recommendation batch in a concise, human-readable story that increases transparency, trust, and click-through rates while satisfying brand tone guidelines. The system can demonstrate awareness of both the user’s history and the underlying rationale for item selection.

System architecture and agent setup. The explanatory workflow is realised as $\text{MAS}_{\text{expl}} = (\mathcal{A}, \mathcal{E}, \Pi)$ with agent ensemble $\mathcal{A} = \{A_{\text{journey}}, A_{\text{session}}, A_{\text{rec}}, A_{\text{expl}}, A_{\text{eval}}\}$. In this MAS, A_{session} invokes Q over $\Omega^{\text{SEM}} \cup \Omega^{\text{EPI}}$ to extract $\mathbf{u}_t$; A_{journey} inspects the live click stream $(x_1, a_1^{\text{user}}, \dots, x_t)$ to tag the latent intent (e.g. “holiday décor upgrade”). Both embeddings flow to A_{rec} which returns recommendation $\mathbf{r}$. The explanation agent A_{expl} feeds $(\mathbf{r}, \mathbf{u}_t)$ into its LLM, then emits $e_t = \Phi_\psi(\mathbf{r}, \mathbf{u}_t)$. A_{eval} verifies $\text{Consistency}(e_t)$ by checking factual mentions against $\mathbf{r}$ and brand constraints stored in Ω^{PROC} ; if the test fails, it sends a revise message back to A_{rec} and A_{expl} , triggering a second-pass generation.

The protocol matrix C therefore admits the cycle $A_{\text{rec}} \leftrightarrow A_{\text{expl}} \rightarrow A_{\text{eval}}$ with veto capability at the evaluator.

Memory and tool requirements. Short-term windows hold the active dialogue and the latest bundle $\mathbf{r}$. Semantic memory captures enduring preferences (“collects Mickey-Mouse merchandise”); episodic slices recall recent campaigns to avoid repetition; procedural memory stores brand-tone exemplars and banned phrasing. Key tools are illustrated in figure 5.

Benefits and discussion. Ω^{PROC} , Ω^{EPI} provide support for outputting truthful and on-brand explanations, while the modular MAS layout lets one inject new style guides or fairness rules by updating Ω^{PROC} and the evaluator prompt. Moreover, because explanations draw on the same semantic and episodic memories that drive ranking, the narrative naturally evolves with the user's journey, reinforcing a sense of personal rapport and transparency throughout the recommendation lifecycle. In other words, when explanations leverage structured memory, real-time data fetching, and specialized text-generation agents, the resulting narratives are more credible and user-centric, ultimately deepening trust in the platform and encouraging repeated interactions.

Fig. 6 illustrates the broader motivation for grounded recommendation explanations: explanatory grouping can make recommendation surfaces more transparent and easier for users to interpret.

5 Operational Challenges and Evaluation Agenda

Agentic recommender systems inherit challenges from general LLM agents and multi-agent systems, but those challenges become recommender-specific when they affect ranking, exposure, feedback, item evidence, user memory, or marketplace outcomes. We therefore organize the challenge section around two questions: (i) what failure mode is introduced by planning, memory, tools, or multi-agent coordination, and (ii) how should a recommender-system researcher measure it?

5.1 Communication Complexity and Protocol

We use *multi-agent communication protocol* (MCP) to mean the set of message schemas, routing rules, context-sharing rules, state-synchronization assumptions, and provenance requirements by which agents exchange information. In an RS setting, MCP design must carry recommendation-specific metadata, including candidate-set provenance, item attributes, retrieved evidence, policy constraints, privacy labels, timestamps, and confidence estimates. Without these fields, downstream agents may rank or explain items using stale, unsupported, or policy-violating context.

Multi-agent recommendation systems rely on multiple autonomous agents that collaborate to deliver relevant items to users, often by sharing user profiles, item information, and intermediate findings. A robust communication protocol is essential in these settings to ensure that agents can exchange messages, negotiate tasks, and maintain synchronized states. In principle, an MCP should provide a unifying framework that allows agents to share user context, coordinate actions, and exchange relevant observations without ambiguity. In practice, however, weak protocol design leads to performance bottlenecks, misinterpretations, and significant implementation overhead. When each agent or data source uses a bespoke interface or message schema, the system devolves into a patchwork of ad-hoc integrations, making it difficult to scale or ensure correct interpretation of exchanged data. These drawbacks are particularly acute in recommendation systems, where rapid, potentially real-time interactions between agents (user-behavior-tracking agent for instance) demand consistency, low latency, and clear semantics of communication. Now, we will review some of the challenges.

Protocol Standardization. A foundational challenge is agreeing on a standardized MCP that all agents within the ecosystem can adopt. Ideally, a newly introduced agent, potentially developed by a different team or vendor, would connect seamlessly to the multi-agent system by adhering to the same syntactic and semantic rules. In practice, such alignment is difficult to achieve. Existing frameworks (for example, the FIPA Agent Communication Language [60]) demonstrated the complexities inherent to specifying communication structures and semantics. Too much rigidity can limit innovation and domain-specific extensions, whereas too little standardization fails to ensure interoperability. A well-designed MCP must be expressive enough to cover the needs of diverse agents (ranging from user-modeling to explanation evaluation for example) while remaining

sufficiently lightweight that developers can implement it without excessive overhead. Achieving wide adoption also poses governance challenges: different organizations may promote competing standards or tailor them to their internal architectures, which underscores the non-technical barriers to protocol unification.

State Synchronization Under High-Frequency Updates. A second technical hurdle is the requirement to maintain synchronized agent states under rapid, potentially continuous updates. If new user actions, item arrivals, or contextual signals flow into the system at high frequency, agents run the risk of operating on inconsistent snapshots of the environment. A naive approach that broadcasts every minor update to all agents creates a scalability bottleneck, as it can saturate the network and introduce large message queues. Yet deferring updates or aggregating them too aggressively can lead to agents making decisions on outdated information. Let

$$\Delta : \mathcal{E}_t \rightarrow \mathcal{E}_{t+1}$$

denote an environment-update function that transitions the shared environment $\mathcal{E}$ from time t to time $t + 1$. The question becomes how the MCP conveys these state transitions Δ to each agent. Approaches from distributed systems, such as eventual consistency or publish/subscribe event channels, may offer partial solutions, but each implies trade-offs among real-time accuracy, network overhead, and agent-level concurrency control. These design decisions directly impact recommendation quality, user-perceived latency, and the ability to scale across thousands of rapidly evolving data points.

Fault Tolerance and Recovery. Another salient concern is that in any distributed system, individual agents or communication links may fail or degrade without warning. If a crucial agent controlling item retrieval collapses, other agents may stall while waiting for updates, ultimately halting the recommendation flow. Protocol-level fault tolerance mitigates such breakdowns by offering mechanisms for detecting failure (via heartbeat signals or timeouts), rerouting tasks, and resynchronizing state when an agent recovers. These mechanisms can range from simple retries to more robust consensus protocols. However, each increment in fault tolerance introduces additional complexity in both agent design and the MCP itself. A well-engineered approach thus needs to strike a balance between resilience and performance overhead, particularly since real-world recommendation engines can ill afford long downtime or incomplete updates, but also cannot endure significant latency due to over-elaborate fault-detection routines.

Scalability in Decentralized Settings. As the number of specialized agents grows, communication traffic rises—potentially in a many-to-many pattern if the system is fully decentralized. This growth can lead to combinatorial message complexity. A naive peer-to-peer architecture may become infeasible as the system matures, so the MCP must support organized communication topologies or hierarchical roles among agents. For example, certain broker agents might aggregate specific types of updates and relay them to other agents, reducing message duplication. Alternatively, partitioning the item space or user segments among subgroups of agents can localize interactions at the expense of global knowledge. Each architectural choice entails a trade-off in coverage, latency, and potential points of failure. Finding near-linear or, at least, sub-exponential scalability solutions remains a major question for multi-agent recommendation. As more data sources or agent roles are added, the system’s design must ensure that communication and processing overheads remain manageable, and that the user still experiences timely, relevant recommendations.

Security and Privacy in Inter-Agent Communication. Agents in a recommendation system often exchange sensitive user data, ranging from personal profiles to behavioral logs, and proprietary or business (e.g. critical information about items and ranking heuristics). Lacking appropriate security

layers, the MCP risks exposing these details to eavesdropping or tampering. Communication protocols must, at minimum, support encryption and authentication so that data is protected against adversarial access and malicious agents cannot masquerade as legitimate participants. Privacy concerns add further layers of complexity, since not every agent should automatically receive all user details. Agents may need to operate on aggregated or encrypted data in a fashion similar to secure multi-party computation, introducing nontrivial overhead and design challenges. Balancing open collaboration, which is essential for rich multi-agent interactions, with robust security constraints is particularly difficult. If the protocol is too permissive, user trust is jeopardized; if it is too stringent, constructive collaboration is stifled.

Open Research Questions and Future Directions. Several unresolved questions emerge from these challenges. One is how to establish an “inclusive but flexible” standard for MCP that agents can adapt across various domains and use cases without spawning multiple incompatible variants. Another is whether “adaptive synchronization schemes” possibly driven by real-time feedback loops, can address the tension between coherence (ensuring all agents remain in sync) and performance (avoiding communication overload). In addition, reliability under fast-changing recommendation contexts remains an open issue. There is also growing interest in robust fault-tolerance paradigms and decentralized consensus strategies that can be layered on top of the MCP [18], potentially leveraging emerging blockchain or distributed ledger approaches[94]. Finally, integrating “privacy-preserving” features, such as secure enclaves or differentially private updates, stands out as a frontier for bridging user data protection with multi-agent collaboration [26, 74]. Addressing these open problems could considerably advance the utility and reliability of multi-agent recommendation systems, enabling them to grow both in scale and sophistication while retaining clarity, resilience, and trustworthiness in their communication fabric.

For evaluation, the failure modes above are measurable. Protocol health can be tracked through message count and schema-validity rate; context integrity through provenance coverage, which records how often a candidate set or claim arrives with the metadata needed to verify it; synchronization quality through the delay between an environment update and its propagation to dependent agents; and coordination quality through inter-agent agreement rate and the ability to attribute a final error to the agent or message that introduced it.

5.2 Scalability

In the context of multi-agent recommendation systems, “scalability” refers to the ability of the architecture to sustain acceptable performance as the system grows in problem size, data volume, or agent complexity. Formally, let N denote the number of users, M the number of items, and A the number of autonomous agents collaborating to produce recommendations. We say that a system is *scalable* if its key performance metrics (e.g., throughput, latency, and accuracy) remain within acceptable bounds when N , M , or A increase, and if resource utilization (computation, memory, financial cost) grows sub-linearly or at most linearly in these parameters. Concretely, if $L(N, M, A)$ is the latency per recommendation request, scalability demands that $L(N, M, A)$ does not grow exponentially with N , M , or A . Analogously, if $\Phi(N, M, A)$ is the throughput of the system, then $\Phi(N, M, A)$ should degrade gracefully (or improve) rather than collapse under increasing loads. This requirement ensures that adding users, expanding the item catalog, or incorporating additional agents does not render the recommendation service unresponsive or cost-prohibitive.

Latency Considerations: Batch vs. Real-Time. Scalability manifests differently in *batch-processing pipelines* versus *real-time inference* scenarios. In an *offline batch setting*, agents may train or update models periodically (e.g., overnight jobs), emphasizing throughput and the completion time of large data-processing tasks. As N and M grow, or as more agent modules participate (for example, separate

modules for each language or region), total computation can increase significantly, stretching batch windows and risking stale or incomplete updates. Partial solutions include distributing tasks across more compute nodes [28], caching intermediate results [83], or consolidating models to reduce overhead [7]. By contrast, *real-time* recommendation places tight upper bounds on per-request latency (often tens of milliseconds). When a user makes a request (e.g., “Recommend me some items now”), the system must orchestrate multiple agents without exceeding a strict time budget. If the request passes sequentially through A agents, or if communication overhead is high, total latency can explode. Methods such as parallelizing sub-tasks, reducing model size (e.g., quantization or distillation), or precomputing partial results become vital. Balancing *freshness* (using the latest user context in real time) with *responsiveness* (controlling inference overhead) is an ongoing engineering challenge, especially as agent complexity and user concurrency grow.

Communication Overhead and Coordination Bottlenecks. A multi-agent system inherently incurs overhead from inter-agent communication. If each agent A_i must exchange intermediate results or context with other agents A_j , the communication pattern can grow as $O(A^2)$ in the worst case. Even more structured topologies (e.g., staged pipelines) introduce data serialization, network transfer, and synchronization costs that increase with A . In real deployments, these costs may overshadow pure compute time, especially if agents are distributed across different servers or data centers. Synchronization points (e.g., where an agent must wait for multiple upstream agents to finish) can compound the bottleneck: a single slow or overloaded agent holds up the entire recommendation pipeline. The result is that response-time variability spikes as A grows, impairing scalability. Proposed strategies include “selective communication” (agents only communicate with relevant peers), asynchronous event-driven designs, or hierarchical orchestration (where specialized “supervisor” or “broker” agents mediate interactions). While each approach can alleviate overhead, designing a protocol that scales sub-linearly in agent count remains an open problem for multi-agent recommenders.

Computational and Financial Costs. Scalability also has a direct economic dimension. Each agent may represent a separate inference step (e.g., a large transformer for generating textual explanations, a collaborative filter for scoring items), thereby multiplying compute, memory, and storage costs when scaled to millions of users. Maintaining separate agents for many languages or market segments compounds these expenses. Even inter-agent communication consumes CPU cycles and bandwidth that can become costly at scale, especially in cloud environments where data transfer and compute times are billed. As a result, system architects must continually balance performance benefits from specialized agents against the financial overhead of running them all. Real-world solutions include consolidating similar tasks into fewer, more general models, caching partial results to avoid redundant computations, or introducing cost-aware scheduling where certain expensive agents (such as large language models) are only invoked for high-value requests or in batch mode.

Examples and Bottlenecks in Practice. Industrial experience underscores these scalability concerns. Large e-commerce providers may need multi-agent approaches for brand-new product lines or language expansions, only to find that naive per-language replication is prohibitively expensive and leads to model silos with limited synergy. Media streaming services, like Netflix or YouTube, rely on multi-stage ranking pipelines, where each stage refines candidates from the previous one. In principle, adding specialized agents can improve personalization; however, at large user scales, the additional latency and resource costs grow burdensome. Academic prototypes such as MACRec [77] demonstrate how multiple LLM-based sub-agents can collaborate for more accurate recommendations, but in production, running multiple large models in parallel can yield super-linear latency and cost, especially if thousands of user queries arrive simultaneously. In each

case, synchronization overheads, data duplication, or model proliferation frequently emerge as bottlenecks.

Open Challenges and Future Directions. Despite progress, significant open questions remain for achieving robust, efficient scalability in multi-agent, language-based recommender systems:

(i) *Dynamic Agent Coordination:* Adaptive scheduling or routing (deciding which agents to invoke per request) could mitigate latency and cost surges; how to automate such coordination at scale is an ongoing research area [56].

(ii) *Cost-Aware Architectures:* Tools for balancing model accuracy with financial overhead remain nascent. Budget-aware or resource-limited modes may become standard features in large multi-agent systems.

(iii) *Multi-Lingual Model Consolidation:* Designing shared representations or truly multilingual models that preserve local performance while avoiding L -fold duplication is an active challenge, especially for low-resource languages.

(iv) *Observability and Debugging:* As agent count grows, tracing the contribution of each agent in a recommendation pipeline becomes non-trivial. New monitoring frameworks are needed to pinpoint bottlenecks, synchronization issues, or poor data hand-offs.

Addressing these challenges will require interdisciplinary solutions that merge distributed systems principles (for fault tolerance, communication efficiency), machine learning insights (for multilingual and multi-domain models), and operational best practices (for cost optimization and maintainability). As multi-agent architectures gain adoption in real-world recommendation services, resolving these open problems stands to bring significant advances in both the quality of personalization and the scalability of next-generation systems.

Scalability is measured along quality, time, and money jointly. The relevant signals are end-to-end and per-agent latency, throughput, token and API cost, and timeout rate under load. Because adding agents trades cost for quality, the informative summary is not any single number but the quality–latency (or quality–cost) Pareto frontier, which exposes whether an extra agent buys a quality gain large enough to justify its latency and cost. Section 6 instantiates exactly this frontier for a concrete ranking task.

5.3 Hallucination, Memory Drift, and Error Propagation

Let $\mathcal{A} = \{A_1, \dots, A_k\}$ be a set of collaborating LLM-based agents in a recommendation pipeline. Each agent A_i produces a message m_i that is consumed, directly or indirectly, by other agents. Define the boolean predicate $\text{valid}(m_i) = 1$ if m_i is factually correct with respect to an external ground-truth oracle $\mathcal{G}$, and 0 otherwise. Hallucination occurs when $\text{valid}(m_i) = 0$. “Error propagation” arises when there exists a path $A_i \rightarrow A_j$ in the agent-interaction graph such that $\text{valid}(m_i) = 0$ and the downstream message m_j is a deterministic function of m_i , yielding $\text{valid}(m_j) = 0$. The probability that at least one final user-facing output is invalid is

$$\Pr[\exists m_u : \text{valid}(m_u) = 0] = 1 - \prod_{i=1}^k (1 - p_i),$$

where $p_i = \Pr[\text{valid}(m_i) = 0]$ after all verification steps. Minimizing this probability in practice is non-trivial; naive composition of agents tends to increase p_i through cascading dependencies.

Mechanisms of Cascading Error. When agents share a common context buffer (e.g. chain-of-thought transcripts), an erroneous assertion by one agent can enter the shared memory Ω_t and be re-ingested by all subsequent agents:

$$\Omega_{t+1} = \Omega_t \cup \{m_i\}, \quad m_j = A_j(\Omega_{t+1}).$$

Because most LLMs lack calibrated epistemic uncertainty, a fabricated but linguistically confident statement in Ω_{t+1} is typically treated as fact [67]. Human cognitive biases such as “authority” or “conformity” find analogues in LLM agents: empirical studies show that a single persuasive hallucination can shift the distribution of responses from peer agents toward the same error, producing group-level misbelief. [44]

A recommender-specific variant of this mechanism is memory drift. Because an agentic recommender persists user state across turns and sessions, an erroneous or outdated fact written to memory is not a one-time error: it is retrieved and re-applied on every subsequent request until it is corrected. A stale preference, an incorrectly merged constraint, or a privacy-sensitive trace that should have been deleted can therefore contaminate future rankings and explanations long after the originating turn, which is the persistent analogue of the cascading error above.

Mitigation Approaches. Current strategies fall into three categories: (i) *Agent-level verification.* Multi-agent debate, majority voting, and cross-examination frameworks instantiate r redundant agents $\{A_i^{(1)}, \dots, A_i^{(r)}\}$ per logical role and accept a message only if a consensus rule $\mathcal{V}$ is satisfied: $m_i^* = \mathcal{V}(m_i^{(1)}, \dots, m_i^{(r)})$. This reduces but does not eliminate correlated hallucinations [37, 50]. (ii) *Supervisor or moderator agents.* A top-level agent A_{mod} inspects each intermediate output and blocks or edits m_i if $\text{valid}(m_i) = 0$. The difficulty is that A_{mod} is itself an LLM with similar failure modes; recursive oversight may be required [49]. (iii) *Tool-assisted grounding.* Agents invoke external APIs $\mathcal{F}$ (e.g. factual search, calculators, product databases) to verify claims: $m_i = f(\text{LLM}(x), \mathcal{F}(x))$. Empirical evidence shows that interleaving of reasoning and tool calls substantially lowers hallucination rates, although it increases latency. [84]

Open Questions for the RecSys Community. Despite these advances, scalability and completeness remain open. Verification ensembles increase compute cost; tool calls add latency; and moderators share the same epistemic blind spots as base agents. These are sample open questions in this regard.

- (1) **Uncertainty Calibration:** How can LLM agents produce calibrated confidence scores that downstream agents may use to discount or challenge low-certainty statements?
- (2) **Efficient Verification:** What lightweight protocols (selective voting, probabilistic auditing, or adaptive tool calls) can bound error propagation without breaching real-time latency constraints?
- (3) **Formal Guarantees:** Can we derive probabilistic upper bounds on $\Pr[\exists m_u : \text{valid}(m_u) = 0]$ for a given agent graph and verification strategy, analogous to error-correcting codes?
- (4) **Adversarial Robustness:** How can multi-agent recommenders detect and neutralize deliberate prompt injections or knowledge-base poisoning that exploit error-propagation pathways?

These failure modes are measurable. Hallucination at the item level is captured by the unsupported-claim rate, citation accuracy, and contradiction rate of agent messages, and by evidence coverage, the fraction of claims backed by retrieved item or user evidence. Memory drift is captured by memory precision and recall against the true user state, by a staleness rate that counts retrieved facts no longer valid, and by deletion compliance, the fraction of removal requests honored on subsequent retrieval.

Answering these questions will be pivotal for building multi-agent, LLM-powered recommenders that harness generative flexibility [13, 14] while safeguarding against cascading hallucinations and the attendant erosion of user trust.

5.4 Potential Collusion or Unintended Emergent Behavior

Multi-agent systems built on large language models (LLMs) are being actively explored for recommendation tasks, enabling multiple autonomous agents to interact and jointly deliver personalized

suggestions. Each agent may correspond to a user perspective, an item provider, or a specialized recommender component, with interactions coordinated through a communication protocol. While this paradigm shows promise for richer, more adaptive recommendations, it also raises concerns over unintended emergent behaviors. When agents possess partially independent objectives and adaptive capabilities, local decision-making can yield global outcomes that system designers did not anticipate. Two key risks are collusion among agents (secretly coordinating strategies that undermine fairness or efficiency) and reinforcing feedback loops that amplify bias or degrade system robustness. This subsection presents current findings and open questions regarding such emergent behaviors, illustrating how they jeopardize fairness, reliability, and user trust in multi-agent LLM-based recommendation systems.

Unintended Emergent Behaviors in Autonomous LLM Agents. “Emergent behavior” in multi-agent LLM systems arises from the complex interactions of multiple agents pursuing local goals, resulting in global patterns unintended by the system’s designers. One salient risk is “collusion”: autonomous agents may discover covert means to collaborate—often through cryptic or hidden messaging—thereby subverting oversight. Prior work has shown that even simple negotiation bots can invent private “shorthand” languages to maximize joint gains, a harbinger of more advanced forms of covert cooperation in LLM-driven systems [36]. Under adversarial incentives, such covert communication can manifest as “cartel-like” behavior, where agents collude to manipulate item rankings or artificially inflate engagement metrics, undermining the recommender’s integrity. In less adversarial but still dynamic settings, agents may spontaneously cooperate if cooperation yields better local utility. For example, self-interested LLM-based “firm” agents in a simulated market might tacitly agree to price-fix, converging to above-competitive prices through repeated interactions—even though no explicit collusion protocol was programmed.

Another persistent concern is bias amplification and feedback loops. Classic recommender systems already exhibit feedback loops where popular items receive even greater exposure, reinforcing their dominance at the expense of niche content [2]. In a multi-agent LLM context, these loops may intensify: a user-simulator LLM’s positive feedback can encourage the recommender LLM to propose narrower sets of items, fostering echo chambers. Empirical explorations reveal that, without careful calibration, multi-agent LLM recommenders can systematically favor popular content or align feedback to confirm preconceived user preferences [41], resulting in reduced diversity and potential “filter bubble” effects. [70]

Case Studies and Examples in LLM-Based Recommendation Systems. While large-scale real-world instances remain relatively uncommon, several research prototypes and industry analogs illustrate emergent misbehaviors:

- (i) *Rec4Agentverse and Similar Frameworks:* Conceptual models wherein multiple Item Agents (each representing a specific content source) coordinate with a central Recommender Agent can inadvertently allow item providers to form alliances, thereby boosting each other. Researchers note the need for robust communication constraints and fairness safeguards to prevent collusive behaviors. [90]
- (ii) *User–Recommender Co-Adaptive Simulations:* Prototypes in which a user-simulator LLM interacts iteratively with a recommender LLM demonstrate how naive feedback loops amplify popularity bias or lead to trivial “approval” behaviors. Even if no intentional collusion exists, the system’s closed feedback cycle can systematically degrade diversity or realism in user modeling. [70]
- (iii) *Industry Observations (Non-LLM but Analogous):* Large platforms like Netflix and Facebook historically reported polarization and echo chambers emerging from automated feedback

loops. Translating these patterns to a multi-agent LLM scenario, if item-provider agents focus excessively on engagement signals, sensational or polarizing content might dominate—a dynamic form of self-reinforcing bias. [55, 73]

5.4.1 Open Challenges. Addressing collusion and unintended emergent behavior in multi-agent LLM recommenders requires an interdisciplinary approach combining AI safety, mechanism design, and robust system engineering. Several key areas demand further exploration:

(i) *Collusion Detection and Prevention.* Identifying covert communication among LLM-based agents remains challenging, particularly if the agents develop coded protocols. Future research might involve adversarial training of oversight models or cryptographic constraints limiting an agent’s ability to conceal signals.

(ii) *Alignment of Local and Global Objectives.* A central cause of emergent misbehavior is the mismatch between local agent incentives and the broader system goal. Mechanism design, incentive-compatible rewards, or hierarchical constraints may help ensure that individually rational actions coincide with socially desirable outcomes.

(iii) *Bias Mitigation in Multi-Agent Loops.* As agents co-adapt, biases can be amplified through cyclical feedback. Tools like causal intervention, re-weighted training, or fairness-promoting prompts must be extended to multi-agent contexts. Evaluating long-term fairness, beyond single-step recommendations, remains an open problem, necessitating new metrics and simulation platforms.

(iv) *Dynamic Adaptation and Safety.* As multi-agent LLM recommenders evolve over time, continuous oversight is required to preempt emergent failures. Online learning, with real-time detection of harmful collusive patterns, will likely be necessary. Balancing adaptability with safety constraints remains a core tension.

By addressing these questions, the recommender systems community can help shape multi-agent LLM architectures that realize their full potential for personalized, adaptive user experiences while safeguarding against collusion, bias, and other emergent hazards. Meeting these challenges will require new techniques in distributed AI governance, fairness-aware design, and robust multi-agent learning—an exciting interdisciplinary frontier for future research.

Measuring these behaviors requires signals beyond per-request accuracy. Collusion can be probed with collusion indicators and price or exposure anomalies under adversarial simulation, and with an objective-conflict rate that quantifies how often local agent incentives diverge from the system objective. Exposure feedback loops are tracked with catalog coverage, long-tail exposure, and exposure disparity across providers or groups, together with group-wise utility, since fairness here is a property of the distribution of exposure over time rather than of a single ranked list.

5.5 Brand, Policy, and Legal Compliance

In agentic recommender systems, brand and policy compliance is not only a generic text-generation issue. It directly affects recommendation explanations, item presentation, generated claims about item attributes or availability, provider obligations, user safety, and legal or regulatory requirements. A policy-checking agent therefore evaluates not only whether the language is on-brand, but also whether the recommended items, claims, evidence, and constraints are consistent with platform policy and applicable law.

Let $\mathcal{A} = \{A_1, \dots, A_k\}$ be a set of collaborating LLM-based agents that generate, transform, or filter textual recommendations. Let $\mathcal{P}$ denote a formal *brand policy*, a finite set of constraints on tone, vocabulary, factual claims, and legally compliant disclosures. For any agent A_i emitting a

textual message $m \in \Sigma^*$, define a *compliance predicate*

$$C_{\mathcal{P}}(m) = \begin{cases} 1 & \text{if } m \text{ satisfies every rule in } \mathcal{P}, \\ 0 & \text{otherwise.} \end{cases}$$

A multi-agent recommender maintains *brand consistency* if, for every output m observable by the end user and for every intermediate message exchanged among agents that could influence m , we have $C_{\mathcal{P}}(m) = 1$. The challenge is to guarantee this condition while preserving the generative flexibility and efficiency of the agents.

Sources of Inconsistency. First, pretrained LLMs embed broad “world knowledge” that may conflict with brand-specific language or regulations. Second, heterogeneous alignment across agents induces drift: if A_1 is fine-tuned for tone but A_2 merely prompted, their joint output may diverge from $\mathcal{P}$. Third, generic safety filters rarely cover nuanced corporate rules (e.g., prohibitions on competitor references, regional marketing laws), leaving gaps through which policy-violating content may pass.

Illustrative Incidents. Well-publicised failures, such as Character.AI’s AI-powered assistant providing harmful advice to a minor, or historical “bomb-making” item bundles on Amazon underline the reputational and legal risks of insufficient control. [68, 69] While these cases did not involve full multi-agent architectures, they foreshadow the compounded risk when multiple autonomous LLMs exchange information without a unifying compliance layer.

Current Mitigation Strategies. Industry is moving toward *brand-tuned LLMs*, in which a base model is fine-tuned or instruction-aligned with proprietary style guides, product catalogs, and regulatory constraints [1]. Researchers have proposed alignment studios and inference-time *policy-expert agents* that veto or rewrite non-compliant tokens:

$$m^* = \arg \max_m \left[\Pr(m \mid \text{context}) \text{ s.t. } C_{\mathcal{P}}(m) = 1 \right].$$

Such approaches reduce manual review but introduce computational overhead and still lack formal guarantees of completeness, especially as $\mathcal{P}$ evolves over time.

Outstanding Challenges. Formal certification of $C_{\mathcal{P}}$ across an entire agent pipeline remains open; policy drift is likely as models and guidelines change asynchronously. Real-time enforcement demands fast, differentiable approximations of $C_{\mathcal{P}}$, yet brand policies frequently involve non-differentiable, context-dependent rules (e.g., comparative advertising limitations differing by jurisdiction). Finally, maintaining multilingual consistency is difficult: a compliant English output may translate into a culturally inappropriate phrase in another language, violating $\mathcal{P}$ ’s spirit even if literal rules are met.

Open Questions for the RecSys Community.

- (1) **Formal Guarantees:** How can we design verifiable protocols or certified decoders that ensure $C_{\mathcal{P}}(m) = 1$ for every agent message without excessive latency?
- (2) **Continuous Compliance:** How can a recommender adapt when either the brand policy or external regulations change, ensuring that legacy agent behaviors do not drift out of compliance?
- (3) **Cross-Lingual Control:** Which multilingual alignment techniques best propagate tone, legal constraints, and cultural sensitivities across languages without retraining separate agents per locale?
- (4) **Evaluation Benchmarks:** What standardized datasets and metrics can quantify brand-policy adherence and stylistic consistency in multi-agent recommendation settings?

Table 4. Operationalizing the five challenges of agentic recommender systems (Sections 5.1–5.5). Each challenge is tied to its source in agentic RS, its recommender-specific manifestation, and measurable evaluation signals.

Challenge	Source in agentic RS	RecSys-specific manifestation	Metrics / protocols
Communication and protocol (§5.1)	Agents exchange candidate sets, evidence, constraints, critiques, and memory records	Stale candidates passed downstream; missing item provenance; unverifiable claims or constraints	Message count; schema-validity rate; provenance coverage; synchronization delay; agreement rate; failure attribution
Scalability and cost (§5.2)	Each agent, memory op, retrieval, tool call, or evaluator adds inference cost and latency	Extra agents improve ranking marginally while exceeding real-time latency budgets	End-to-end and per-agent latency; token / API cost; throughput; time-out rate; quality-latency Pareto frontier
Hallucination and memory drift (§5.3)	Generated claims, retrieved evidence, and stored memories are consumed downstream as if factual	Unsupported attributes; fabricated availability; stale preferences; invalid constraint satisfaction; privacy-sensitive recall	Unsupported-claim rate; citation accuracy; contradiction rate; evidence coverage; memory precision/recall; staleness rate; deletion compliance
Collusion and exposure loops (§5.4)	Autonomous agents optimize local goals conflicting with user, provider, or fairness objectives	Provider agents suppress competitors; engagement over-optimization; simulated users amplify popularity bias	Collusion indicators; price / exposure anomalies; objective-conflict rate; catalog coverage; long-tail exposure; exposure disparity; group-wise utility
Brand, policy, and legal compliance (§5.5)	Explanations, claims, and actions are generated dynamically and vary across users, markets, and languages	Off-brand tone; prohibited or unsafe claims; unsupported legal/product claims; policy-inconsistent justifications	Guideline adherence; policy-violation rate; cross-lingual consistency; human-review agreement; escalation rate

- (5) **Cost–Benefit Trade-offs:** How do we balance the computational and financial overhead of stringent compliance layers against their risk-mitigation benefits in large-scale production systems?

Addressing these questions will be pivotal for deploying multi-agent, LLM-based recommenders that are both creative and rigorously aligned with brand identity and regulatory obligations. Operationally, compliance is measured by guideline-adherence and policy-violation rates over generated outputs, cross-lingual consistency of the same recommendation across locales, human-review agreement on flagged cases, and the escalation rate at which the system defers to human oversight.

Table 4 operationalizes these presented challenges by mapping each challenge to its source in agentic recommendation, its recommender-specific manifestation, and measurable metrics or protocols.

6 Empirical Illustration: When Do Multi-Agent Pipelines Pay Off?

The preceding sections argued, conceptually, that agentic orchestration unlocks capabilities beyond single-pass pipelines. A perspective is more convincing when it is made concrete, and in this section we want to quantify both the benefit and the cost of multi agentic structures. This section therefore reports a controlled study on the simplest slice of the Interactive Recommendation task of Section 4.1: re-ranking a fixed candidate set based on a user’s purchase history. The study is deliberately narrow so that the effect of *adding agents* can be isolated from confounds such as multi-turn dialogue dynamics or tool availability.

In this section, we want to explore what makes these orchestrations work and when they will not benefit the ultimate goal of the task. We will see for the given task of conversational ranking, the value of using multi agentic orchestrations scales with the complexity of the input. We will also show that several of the multi-agentic roles that look helpful in principle do not repay their cost in

practice. For instance, we show that self-referential refinement can actively degrade ranking quality. Concretely, this study provides controlled evidence for the usage of multi-agentic mechanisms. Throughout, we connect the measurements back to the error-propagation formalism of Section 5.3 and the scalability concerns of Section 5.2.

6.1 Experimental Setup

Task. Each instance is a single ranking question: given the purchase history of a user and ten candidate items, the pipeline must return a ranking of the ten candidates. Exactly one candidate is the held-out next purchase; the remaining nine are sampled at random from items in the same product category. The agent emits a ranked list, and specialized agents cooperate to produce it.

Data. We use the Amazon-2023 review corpus [30] across four categories: Amazon Fashion, Appliances, Electronics, and Toys and Games. For each category we draw two contrastive user cohorts of 100 users, a *random* cohort and a *high-diversity* cohort, giving two samples of $n = 400$ pooled over the four categories (for detailed per category results see Appendix A). The contrast between these two samples is the backbone of the analysis.²

To assess how recommendation quality varies with the heterogeneity of a user’s interaction history, we stratify users by the lexical diversity of the item titles they have previously engaged with. For a user u with history $H_u = \{t_1, \dots, t_m\}$, we represent each item title t_i by its set of lowercased word tokens $T_i = \text{tokens}(t_i)$, and define the diversity of u as the mean pairwise Jaccard distance over all title pairs in the history:

$$\text{Div}(u) = \binom{m}{2}^{-1} \sum_{1 \leq i < j \leq m} \left(1 - \frac{|T_i \cap T_j|}{|T_i \cup T_j|} \right), \quad (2)$$

where $\binom{m}{2}$ is the number of unordered title pairs. The score lies in $[0, 1]$: it is 0 when all titles share an identical token set (a homogeneous history) and approaches 1 when every pair of titles is lexically disjoint (a maximally diverse history). Users with fewer than two non-empty token sets are excluded, and histories are capped at 100 titles to bound the quadratic pairwise cost.

We restrict the analysis to users with at least seven history items. The *high-diversity* cohort comprises, per category, the 100 users with the largest $\text{Div}(u)$. The *random* cohort is drawn uniformly at random (seed 42) from the eligible users in the same category, and serves as a typical baseline population against which the high-diversity cohort is compared.

Agentic roles and workflows. We evaluate seven workflows, summarized in Table 5, grouped into three families of agentic role. The single-shot LLM baseline (SA) issues one model call that reads the history and the candidates and returns the ranking, and serves as the reference against which the value of additional agents is measured. The *decomposition and specialization* family restructures the input before ranking: a profiler compresses the history into an intent summary that the ranker then conditions on (PR), and a planner additionally outlines the ranking strategy (PPR). This family operationalizes the planning and task-decomposition capability that breaks a goal into focused subtasks each solved by a dedicated call [46, 61]. The *ensembling and aggregation* family replaces the single ranker with several rankers sampled at higher temperature for diversity and reconciles them through an arbitrator, either conditioned on a plan alone (PENs) or on both a plan and a profile (PPENs); this follows the finding that sampling multiple agents and aggregating their outputs improves accuracy, with gains that grow with task difficulty [46]. The *iterative and adversarial refinement* family improves an initial ranking through feedback rather than through added input

²All pipeline implementations, prompt templates, the sampling scripts, and the scripts that regenerate every results table in this paper are available at <https://github.com/RezaYM/agenticrecsys.git>.

Table 5. Agentic workflows evaluated, grouped by role family. Calls/q is the number of LLM invocations per query; the original implementation identifiers are given in parentheses.

Workflow (ID)	Role family	Calls/q
SA: Single-shot LLM call	baseline	1
PR: Profiler → Ranker	decomposition / specialization	2
PPR: Planner+Profiler → Ranker	decomposition / specialization	3
PENS: Planner → 3 Rankers → Arbitrator	ensembling / aggregation	5
PPENS: Planner+Profiler → 3 Rankers → Arbitrator	decomposition + ensembling	6
RC: Ranker ↔ Critic	iterative / adversarial	3
DEB: Two-agent Debate	iterative / adversarial	4

structure: a ranker is critiqued by an evaluator and then revises (RC), in the manner of single-model self-feedback loops [47], or two rankers exchange and revise their rankings over several turns of debate (DEB) [15].

Model, metrics, and cost basis. All pipelines use gpt-5-mini-2025-08-07. We evaluate ranking quality with MRR, NDCG, and HitRate (equal to Recall) at cutoffs $k \in \{3, 10\}$. Cost is reported as input/output tokens, US dollars at \$0.25/\$2.00 per million input/output tokens, and the number of LLM calls per query.

6.2 Quantifying the Cost of Agentic Structures

Before asking whether added agents improve ranking, we fix the bar that any quality gain must clear. Table 6 reports per-query tokens, dollar cost, and latency for both samples. The single shot LLM call issues one call of roughly 1.7k tokens at about \$0.0022 per query. The ensemble workflows PENS and PPENS issue five and six calls, consume between 8.9k and 10.8k tokens, and cost between \$0.0112 and \$0.0123 per query, which is between five and six times the Single-shot LLM call cost.

We report latency along the parallel critical path rather than the raw sequential sum, because the sequential figure is an artifact of our single-threaded benchmark harness rather than a property of the workflow’s dependency structure. The critical path is the longest chain of calls that must run in series once mutually independent calls are dispatched concurrently. Take PENS as an example: the planner runs first, its plan is handed to three rankers that are mutually independent and can therefore be dispatched at the same time, and the arbitrator runs last on their combined output. The critical path is thus planner, then one ranker stage (not three), then arbitrator, so the five-call workflow has the latency of three calls in series rather than five. On the high-diversity sample this places PENS and PPENS at roughly three times the Single-shot LLM call latency (34.1 s and 35.2 s against 11.8 s), and somewhat higher on the random sample, where the single call is itself slower. Crucially, parallel execution recovers a share of the wall-clock penalty but does not reduce token consumption or dollar cost, which scale with the number and size of calls regardless of how they are scheduled. The implication for the rest of the section is simple: a workflow earns its place only if its quality improvement is large enough to justify a multiplicative increase in cost.

6.3 On Typical Users, Added Agents Do Not Pay, and Often Hurt

We begin with the negative result, since it is the more surprising of the two and frames the rest of the section. Table 7 reports quality on the random sample. On these typical histories the single shot LLM call is already strong, reaching an NDCG@10 of 0.7197, and no multi-agent pipeline meaningfully improves on it. The best multi-agent score, 0.7209 from the debate pipeline, is within a fraction of a point of the baseline, and the two pipelines tie on NDCG@3 as well. Given the

Table 6. Cost and latency per query on gpt-5-mini, aggregate over the four categories ($n = 400$). “Total tok/q” sums input and output tokens over a workflow’s calls. “Lat. par” is the parallel critical-path latency, defined as the longest chain of dependent calls once mutually independent calls are dispatched concurrently. Lower is better throughput; the single shot LLM call is the floor on every column by construction.

Workflow	Calls/q	Total tok/q	Cost/q (USD)	Lat. par (s)
<i>Random sample ($n = 400$)</i>				
SA: Single-shot LLM call	1.0	1,756	0.00217	14.52
PR: Profiler $\rightarrow$ Ranker	2.0	2,804	0.00328	22.31
PPR: Planner+Profiler $\rightarrow$ Ranker	3.0	4,273	0.00473	22.98
RC: Ranker $\leftrightarrow$ Critic	3.0	6,699	0.00831	55.27
DEB: Two-agent Debate	4.0	7,635	0.00933	62.92
PENS: Planner $\rightarrow$ 3 Rankers $\rightarrow$ Arb.	5.0	9,510	0.01121	38.36
PPENS: Planner+Profiler $\rightarrow$ 3 Rankers $\rightarrow$ Arb.	6.0	10,812	0.01225	44.56
<i>High-diversity sample ($n = 400$)</i>				
SA: Single-shot LLM call	1.0	1,654	0.00219	11.76
PR: Profiler $\rightarrow$ Ranker	2.0	2,537	0.00322	17.70
PPR: Planner+Profiler $\rightarrow$ Ranker	3.0	3,912	0.00470	20.85
RC: Ranker $\leftrightarrow$ Critic	3.0	6,221	0.00803	41.89
DEB: Two-agent Debate	4.0	7,130	0.00923	49.61
PENS: Planner $\rightarrow$ 3 Rankers $\rightarrow$ Arb.	5.0	8,921	0.01119	34.14
PPENS: Planner+Profiler $\rightarrow$ 3 Rankers $\rightarrow$ Arb.	6.0	10,000	0.01202	35.19

five- to six-fold cost increase documented in Table 6, this is effectively a loss: the added structure purchases no useful signal.

More striking is that two pipelines degrade quality. The Ranker+Evaluator pipeline (RC) falls to an NDCG@10 of 0.6922, a relative drop of 3.8 percent against the single shot LLM call, and a larger 7.2 percent drop at NDCG@3, while costing roughly four times as much. We read this as a direct empirical instance of the error-propagation mechanism formalised in Section 5.3. When the evaluator critiques an already-correct ranking with no external ground-truth oracle, its critique is itself an ungrounded LLM output that the ranker then incorporates, so a second source of error is introduced where the first pass had none. In the notation of that section, $\Pr[\exists m_u : \text{valid}(m_u) = 0] = 1 - \prod_i (1 - p_i)$ grows when an additional stage with $p_i > 0$ is composed onto a pipeline that already produces a good answer. The cleanest summary of this subsection is that on inputs a single model already handles well, additional agents add failure surface faster than they add signal.

6.4 On High-Diversity Users, Decomposition and Ensembling Begin to Pay

The picture inverts on the high-diversity sample. Table 8 shows that the absolute scores are lower than on the random sample, which is expected because diverse histories are harder to summarise into a single purchase intent. The informative quantity, however, is the relative ordering: here the pipelines that carry a planner or an ensemble lead the single shot LLM call. At NDCG@3 the combined pipeline (PPENS) reaches 0.5202 and the ensemble pipeline (PENS) reaches 0.5162, against 0.4898 for the single shot LLM call, a relative gain of 6.2 and 5.4 percent respectively. At NDCG@10 the ensemble pipeline leads at 0.6471 against 0.6316, a gain of 2.5 percent, and the MRR columns tell the same story, with PENS best at every reported cutoff.

The contrast between Tables 7 and 8 is the central empirical finding of the paper: the same pipelines that are wasteful on typical users become beneficial on diverse ones. This is consistent

Table 7. Ranking quality on the random sample, aggregate over four categories ($n = 400$), gpt-5-mini. Best and second-best among the LLM pipelines per column are **bold** and underlined. Higher is better.

Method	MRR@3	MRR@10	NDCG@3	NDCG@10
SA Single-shot LLM call	<u>0.5846</u>	0.6317	<u>0.6225</u>	0.7197
RC Ranker+Evaluator	0.5400	0.5960	<u>0.5779</u>	0.6922
PR Profiler+Ranker	0.5775	0.6255	0.6191	0.7156
PPR Planner+Profiler+Ranker	0.5708	0.6185	0.6129	0.7100
PENS Planner+3 Rankers+Arb.	0.5792	0.6266	0.6190	0.7159
PPENS Planner+Profiler+3 Rankers+Arb.	0.5837	<u>0.6319</u>	0.6212	<u>0.7198</u>
DEB Two-agent Debate	0.5850	0.6330	0.6229	0.7209
Random baseline	0.1833	0.2929	0.2131	0.4544

Table 8. Ranking quality on the high-diversity sample, aggregate over four categories ($n = 400$), gpt-5-mini. Best and second-best among the LLM pipelines per column are **bold** and underlined. Higher is better.

Method	MRR@3	MRR@10	NDCG@3	NDCG@10
SA: Single-shot LLM call	0.4492	0.5174	0.4898	0.6316
RC: Ranker+Evaluator	0.4379	0.5084	0.4788	0.6247
PR: Profiler+Ranker	0.4537	0.5243	0.4919	0.6368
PPR: Planner+Profiler+Ranker	0.4617	0.5272	0.5069	0.6401
PENS: Planner+3 Rankers+Arb.	0.4725	0.5366	<u>0.5162</u>	0.6471
PPENS: Planner+Profiler+3 Rankers+Arb.	<u>0.4704</u>	<u>0.5290</u>	0.5202	<u>0.6416</u>
DEB: Two-agent Debate	0.4533	0.5215	0.4946	0.6349
Random baseline	0.1833	0.2929	0.2131	0.4544

with the intuition that decomposition and aggregation only have something to contribute when the input is heterogeneous enough that a single pass under-resolves it.

6.5 Role-by-Role Analysis: What Each Agent Contributes

Reading the two quality tables by role family clarifies which forms of agentic structure carry the high-diversity gain and which do not.

Decomposition and specialisation. The planner and the profiler are the source of the steady, low-variance improvement on diverse inputs. Adding the profiler alone (PR) moves NDCG@10 from 0.6316 to 0.6368 on the high-diversity sample while leaving it slightly below the baseline on the random sample (0.7156 against 0.7197). Adding a planner on top (PPR) lifts NDCG@3 to 0.5069. The mechanism is intuitive: when a history spans many interests, articulating a ranking strategy and compressing the history into an explicit intent both reduce the burden on the final ranking step. When the history is already coherent, there is little to decompose, and the extra calls neither help nor hurt the ranking while still costing tokens. This is the empirical content of capability (i), planning and task decomposition.

Ensembling and aggregation. The largest high-diversity gains come from the ensemble pipelines PENS and PPENS, which lead at NDCG@10 and NDCG@3 respectively. Sampling several rankers at elevated temperature and reconciling them with an arbitrator recovers signal that any single sample

misses, which matters precisely when the input is ambiguous. This is also the most expensive family, so its advantage is real but costly.

Iterative and adversarial refinement. This family is the weakest. The evaluator pipeline (RC) is the worst performer on the random sample and remains below the baseline on the high-diversity sample, and the debate pipeline (DEB) is roughly neutral on both. The synthesis is that the gains observed in this study come from restructuring the input and aggregating diverse views, not from agents critiquing one another in a closed loop. Closed-loop critique without an external oracle tends, if anything, to introduce the cascading errors analyzed in Section 5.3.

6.6 The Profiler as Memory: Why Compression Helps on Diverse Histories

The profiler can be read through the memory formalism of Section 2.3. It instantiates the retention operator $\mathcal{R}$ of Definition 2.3 : it distills the raw history into a compact intent summary $\tilde{C}$, on which the ranker then conditions through the retrieval operator $\mathcal{Q}$ of Definition 2.4. We are explicit that this is working-memory, or session-scoped, compression.

The evidence for the value of this memory mechanism is the profiler-only pipeline (PR). On the high-diversity sample it improves NDCG@10 from 0.6316 to 0.6368 and MRR@10 from 0.5174 to 0.5243, whereas on the random sample it sits marginally below the single shot LLM call (0.7156 against 0.7197 at NDCG@10). In other words, compressing a history into an explicit intent helps exactly when the history is rich enough to make compression informative, and is otherwise a small, avoidable expense.

7 Conclusion

This paper established a footing for agentic recommender systems: architectures in which stateful agents, memories, tools, communication protocols, and verification steps are composed to improve recommendation-layer outcomes. We began by formalizing the core building blocks, generic LLM agents, recommender agents, multi-agent systems, memory update and retrieval functions, and observable traces, and by bounding what is recommendation-specific in such a system. These abstractions unify design choices such as raw buffers, vector stores, knowledge graphs, procedural memories, and tool-mediated evidence retrieval into a vocabulary that is precise enough for evaluation while remaining implementation-flexible.

Building on this vocabulary, Section 5 surfaced the operational fault lines that accompany such flexibility. We organized the open problems into five challenge families, communication complexity and protocol design, scalability and cost, hallucination and error propagation, emergent misalignment and collusion, and brand, policy, and legal compliance, and connected each to recommender-specific manifestations and measurable signals such as ranking quality, evidence coverage, provenance and trace validity, latency, token cost, exposure disparity, policy-violation rate, and judge reliability. The recurring message is that progress depends not only on larger models but on the interaction rules, memory hierarchies, and incentive structures that govern how agents are composed.

Finally, the controlled study of Section 6 shows why agentic recommendation must be evaluated conditionally. More agents are not automatically better: on representative next-item ranking samples the single-agent baseline is the pareto efficient default, whereas decomposition and ensemble roles become useful for high-diversity user histories, and closed-loop self-criticism can even degrade ranking quality. This supports a practical design principle in which agentic complexity is routed to the inputs where its marginal quality gain justifies its added latency, cost, and governance risk, and an evaluation agenda in which that trade-off, rather than top- K accuracy alone, is what future work should report.

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A Per-Category Empirical Results

Section 6 reported ranking quality and cost pooled over the four Amazon-2023 categories. This appendix disaggregates those pooled numbers into the individual categories (*Amazon Fashion*, *Appliances*, *Electronics*, and *Toys and Games*; $n=100$ users per category and cohort) for both the random cohort (Tables 9 and 11) and the high-diversity cohort (Tables 10 and 12). Workflow identifiers follow Table 5; in every category block the best and second-best values per column among the seven LLM pipelines are shown in **bold** and underline, and the random-ranking baseline is listed once per table. Because each user has a single held-out positive, $\text{MAP}@k = \text{MRR}@k$; we therefore report MRR (equivalently MAP) and NDCG, and include the MAP columns for completeness. The two headline findings of the main text, namely that single-shot ranking is hard to beat on typical users (Table 7) and that decomposition and ensembling pay off on diverse users (Table 8), both survive disaggregation, but the per-category view shows that each effect is unevenly distributed across categories.

A.1 Ranking Quality by Category

Random cohort. Table 9 shows that the pooled near-tie between the single shot LLM call and the best multi-agent pipeline conceals substantial category heterogeneity. On *Electronics* the single shot LLM call (SA) is strictly best on all four reported cells, and the evaluator pipeline RC collapses to an $\text{NDCG}@10$ of 0.6602, about 8.6% below SA's 0.7226; this is the clearest single-category instance of the error-propagation effect of Section 5.3. On the other three categories a multi-agent pipeline does edge ahead of SA, but the gains are small: PPENS leads on *Toys and Games* ($\text{NDCG}@10$ 0.7491 vs. 0.7313, about +2.4%) at roughly six times the cost, DEB leads on *Amazon Fashion* (0.7364 vs. 0.7241, about +1.7%) at roughly four times the cost, and the inexpensive PR leads on *Appliances* (0.7114 vs. 0.7007, about +1.5%) at only about 1.5 $\times$ the cost. Only the last of these is plausibly cost-justified, and even it is a sub-2% gain. The evaluator pipeline RC is the weakest LLM pipeline in every category, so the pooled conclusion that SA is Pareto-efficient on typical users is driven substantially by *Electronics*, with the remaining categories contributing only small, cost-inefficient multi-agent gains.

High-diversity cohort. Table 10 confirms that the multi-agent advantage on diverse histories holds in every category, but with a magnitude that tracks how much room the single shot LLM call leaves. The gains are largest on the categories where SA struggles most: on *Amazon Fashion*, PENS lifts $\text{NDCG}@3$ from 0.4323 to 0.5064 (about +17%) and $\text{NDCG}@10$ from 0.5943 to 0.6341 (about +6.7%); on *Appliances*, PPENS lifts $\text{NDCG}@3$ from 0.4886 to 0.5401 (about +10.5%) and PENS lifts $\text{NDCG}@10$ by about 3.9%. On *Electronics* the ensembling and decomposition pipelines lead more modestly (PENS $\text{NDCG}@10$ 0.6389 vs. 0.6266, about +2.0%). *Toys and Games* is the boundary case: it has the highest single-agent baseline of the four categories, so there is little to recover, and the pure ensemble PENS actually dips below SA at $\text{NDCG}@10$ (0.6628 vs. 0.6772) while only DEB edges ahead (about +0.5%). This matches the paper's mechanism: agentic restructuring helps when a single pass under-resolves the input, and a high-diversity history in an otherwise easy category does not always under-resolve it. RC is again the weakest pipeline throughout.

A.2 Cost and Efficiency by Category

Tables 11 and 12 report input, output, and total tokens per query, parallel critical-path latency, and dollar cost per category. The ordering is identical in every category: the single shot LLM call is the floor on every column, the profiler-ranker pipeline PR is second, and the ensembles PENS and PPENS cost five to six times the single-agent baseline. The absolute cost, however, tracks history length and therefore varies by category: *Electronics* is the most token-heavy (random SA totals

Table 9. Per-category ranking quality on the *random* cohort (gpt-5-mini, $n=100$ per category). Best and second-best per column among the LLM pipelines are **bold** and underlined; higher is better. Because each user has one held-out positive, $\text{MAP}@k = \text{MRR}@k$, so the MAP columns repeat the MRR columns and are included only for completeness.

Method	MRR		NDCG		MAP	
	@3	@10	@3	@10	@3	@10
<i>Amazon Fashion (n=100)</i>						
SA	0.5850	0.6380	0.6170	0.7241	0.5850	0.6380
RC	0.5683	0.6224	0.5996	0.7118	0.5683	0.6224
PR	0.5583	0.6156	0.5946	0.7077	0.5583	0.6156
PPR	0.5883	0.6357	0.6273	0.7231	0.5883	0.6357
PENS	<u>0.6017</u>	<u>0.6467</u>	<u>0.6396</u>	<u>0.7313</u>	<u>0.6017</u>	<u>0.6467</u>
PPENS	0.5900	0.6391	0.6259	0.7255	0.5900	0.6391
DEB	0.6067	0.6538	0.6407	0.7364	0.6067	0.6538
<i>Appliances (n=100)</i>						
SA	0.5617	0.6074	0.6023	0.7007	0.5617	0.6074
RC	0.5183	0.5739	0.5599	0.6753	0.5183	0.5739
PR	0.5783	0.6204	0.6223	0.7114	0.5783	0.6204
PPR	0.5467	0.5963	0.5883	0.6923	0.5467	0.5963
PENS	0.5483	0.5971	0.5894	0.6926	0.5483	0.5971
PPENS	<u>0.5733</u>	<u>0.6197</u>	<u>0.6107</u>	<u>0.7097</u>	<u>0.5733</u>	<u>0.6197</u>
DEB	0.5600	0.6085	0.5986	0.7016	0.5600	0.6085
<i>Electronics (n=100)</i>						
SA	0.5917	0.6355	0.6301	0.7226	0.5917	0.6355
RC	0.4967	0.5535	0.5438	0.6602	0.4967	0.5535
PR	0.5717	0.6166	0.6173	0.7087	0.5717	0.6166
PPR	0.5633	0.6074	0.6114	0.7022	0.5633	0.6074
PENS	<u>0.5800</u>	<u>0.6295</u>	<u>0.6186</u>	<u>0.7183</u>	<u>0.5800</u>	<u>0.6295</u>
PPENS	0.5483	0.5985	0.5927	0.6951	0.5483	0.5985
DEB	0.5717	0.6179	0.6151	0.7099	0.5717	0.6179
<i>Toys and Games (n=100)</i>						
SA	0.6000	0.6459	0.6407	0.7313	0.6000	0.6459
RC	0.5767	0.6340	0.6083	0.7216	0.5767	0.6340
PR	<u>0.6017</u>	0.6495	<u>0.6423</u>	0.7345	<u>0.6017</u>	0.6495
PPR	0.5850	0.6347	0.6244	0.7225	0.5850	0.6347
PENS	0.5867	0.6333	0.6283	0.7216	0.5867	0.6333
PPENS	0.6233	0.6704	0.6555	0.7491	0.6233	0.6704
DEB	<u>0.6017</u>	<u>0.6518</u>	0.6370	<u>0.7356</u>	<u>0.6017</u>	<u>0.6518</u>
Random baseline	0.1833	0.2929	0.2131	0.4544	0.1833	0.2929

2,050 tokens and PPENS 12,191), whereas *Toys and Games* is the lightest (high-diversity SA 1,497). The categories on which multi-agent pipelines help most are thus also among the most expensive to run them on, which sharpens rather than softens the cost–benefit tension of Section 6.2. The latency columns make the penalty on iterative refinement concrete: because RC and DEB cannot overlap their calls, they are the slowest pipelines in wall-clock terms even though they issue fewer

Table 10. Per-category ranking quality on the *high-diversity* cohort (gpt-5-mini, $n=100$ per category). Best and second-best per column among the LLM pipelines are **bold** and underlined; higher is better. The MAP columns coincide with the MRR columns (single positive).

Method	MRR		NDCG		MAP	
	@3	@10	@3	@10	@3	@10
<i>Amazon Fashion (n=100)</i>						
SA	0.3917	0.4694	0.4323	0.5943	0.3917	0.4694
RC	<u>0.4317</u>	0.4998	<u>0.4799</u>	0.6189	<u>0.4317</u>	0.4998
PR	0.4200	<u>0.5022</u>	0.4533	<u>0.6196</u>	0.4200	<u>0.5022</u>
PPR	0.4200	0.4948	0.4714	0.6166	0.4200	0.4948
PENS	0.4567	0.5188	0.5064	0.6341	0.4567	0.5188
PPENS	0.4217	0.4917	0.4723	0.6132	0.4217	0.4917
DEB	0.3883	0.4776	0.4196	0.6004	0.3883	0.4776
<i>Appliances (n=100)</i>						
SA	0.4467	0.5128	0.4886	0.6281	0.4467	0.5128
RC	0.4283	0.4938	0.4775	0.6143	0.4283	0.4938
PR	0.4483	0.5189	0.4873	0.6329	0.4483	0.5189
PPR	0.4617	0.5229	0.5104	0.6373	0.4617	0.5229
PENS	0.4883	0.5434	<u>0.5375</u>	0.6527	0.4883	0.5434
PPENS	0.4883	<u>0.5417</u>	0.5401	<u>0.6518</u>	0.4883	<u>0.5417</u>
DEB	<u>0.4650</u>	0.5251	0.5125	0.6383	<u>0.4650</u>	0.5251
<i>Electronics (n=100)</i>						
SA	0.4450	0.5106	0.4899	0.6266	0.4450	0.5106
RC	0.3900	0.4685	0.4259	0.5928	0.3900	0.4685
PR	0.4433	0.5094	0.4886	0.6258	0.4433	0.5094
PPR	<u>0.4550</u>	<u>0.5216</u>	0.4975	<u>0.6352</u>	<u>0.4550</u>	<u>0.5216</u>
PENS	0.4567	0.5259	<u>0.4988</u>	0.6389	0.4567	0.5259
PPENS	0.4533	0.5113	0.5088	0.6285	0.4533	0.5113
DEB	0.4367	0.5016	0.4883	0.6204	0.4367	0.5016
<i>Toys and Games (n=100)</i>						
SA	0.5133	<u>0.5769</u>	0.5483	<u>0.6772</u>	0.5133	<u>0.5769</u>
RC	0.5017	0.5715	0.5320	<u>0.6727</u>	0.5017	0.5715
PR	0.5033	0.5668	0.5383	0.6691	0.5033	0.5668
PPR	0.5100	0.5694	0.5483	0.6715	0.5100	0.5694
PENS	0.4883	0.5582	0.5220	0.6628	0.4883	0.5582
PPENS	<u>0.5183</u>	0.5714	0.5596	0.6729	<u>0.5183</u>	0.5714
DEB	0.5233	0.5819	<u>0.5581</u>	0.6804	0.5233	0.5819
Random baseline	0.1833	0.2929	0.2131	0.4544	0.1833	0.2929

calls than the ensembles. On random *Electronics*, for example, RC (72.1 s) and DEB (69.3 s) both exceed PENS (41.9 s), whose three rankers run concurrently, while SA stays the latency floor at 14.8 s. This reinforces the role analysis of Section 6.5: closed-loop critique is penalized on quality and on latency at the same time.

Table 11. Per-category cost and efficiency on the *random* cohort (gpt-5-mini, $n=100$ per category). “Total” sums input and output tokens; “Lat. par” is the parallel critical-path latency in seconds; cost is in USD at \$0.25/\$2.00 per 1M input/output tokens. Lower is better on every column; best and second-best per column are **bold** and underlined.

Method	In tok/q	Out tok/q	Total tok/q	Lat. par (s)	Cost/q (USD)
<i>Amazon Fashion (n=100)</i>					
SA	612	895	1,507	15.03	0.00194
RC	2,372	3,546	5,918	39.97	0.00768
PR	<u>1,070</u>	<u>1,406</u>	<u>2,477</u>	<u>17.50</u>	<u>0.00308</u>
PPR	<u>1,769</u>	<u>2,047</u>	<u>3,816</u>	<u>20.68</u>	<u>0.00454</u>
PENS	3,685	4,747	8,432	34.39	0.01041
PPENS	4,485	5,186	9,671	40.33	0.01149
DEB	2,766	3,808	6,574	56.88	0.00831
<i>Appliances (n=100)</i>					
SA	769	1,066	1,835	15.03	0.00232
RC	2,959	3,709	6,668	57.73	0.00816
PR	<u>1,320</u>	<u>1,484</u>	<u>2,803</u>	<u>23.16</u>	<u>0.00330</u>
PPR	<u>2,171</u>	<u>2,144</u>	<u>4,315</u>	<u>24.99</u>	<u>0.00483</u>
PENS	4,482	5,524	10,006	39.44	0.01217
PPENS	5,340	5,872	11,212	49.86	0.01308
DEB	3,430	4,611	8,041	69.37	0.01008
<i>Electronics (n=100)</i>					
SA	972	1,078	2,050	14.83	0.00240
RC	3,618	4,136	7,754	72.07	0.00918
PR	<u>1,688</u>	<u>1,624</u>	<u>3,312</u>	<u>29.51</u>	<u>0.00367</u>
PPR	<u>2,754</u>	<u>2,212</u>	<u>4,966</u>	<u>24.82</u>	<u>0.00511</u>
PENS	5,532	5,199	10,731	41.94	0.01178
PPENS	6,594	5,597	12,191	47.89	0.01284
DEB	4,242	4,575	8,817	69.29	0.01021
<i>Toys and Games (n=100)</i>					
SA	704	927	1,632	13.18	0.00203
RC	2,675	3,780	6,455	51.32	0.00823
PR	<u>1,232</u>	<u>1,391</u>	<u>2,623</u>	<u>19.08</u>	<u>0.00309</u>
PPR	<u>2,022</u>	<u>1,974</u>	<u>3,996</u>	<u>21.42</u>	<u>0.00445</u>
PENS	4,162	4,710	8,871	37.64	0.01046
PPENS	5,009	5,163	10,173	40.17	0.01158
DEB	3,134	3,976	7,109	56.13	0.00873

Table 12. Per-category cost and efficiency on the *high-diversity* cohort (gpt-5-mini, $n=100$ per category). Columns and conventions are as in Table 11; lower is better, best and second-best per column in **bold** and underline.

Method	In tok/q	Out tok/q	Total tok/q	Lat. par (s)	Cost/q (USD)
<i>Amazon Fashion (n=100)</i>					
SA	572	944	1,516	11.20	0.00203
RC	2,255	3,520	5,775	39.68	0.00760
PR	<u>990</u>	<u>1,455</u>	<u>2,445</u>	<u>18.39</u>	<u>0.00316</u>
PPR	1,646	2,107	3,753	22.27	0.00462
PENS	3,468	5,055	8,523	37.09	0.01098
PPENS	4,232	5,264	9,496	35.76	0.01159
DEB	2,606	4,009	6,615	48.44	0.00867
<i>Appliances (n=100)</i>					
SA	713	1,143	1,857	12.36	0.00246
RC	2,762	3,683	6,444	42.26	0.00806
PR	<u>1,188</u>	<u>1,505</u>	<u>2,693</u>	<u>17.83</u>	<u>0.00331</u>
PPR	1,978	2,253	4,231	21.95	0.00500
PENS	4,149	5,711	9,860	35.23	0.01246
PPENS	4,964	6,000	10,964	35.62	0.01324
DEB	3,170	4,614	7,784	51.00	0.01002
<i>Electronics (n=100)</i>					
SA	727	1,018	1,745	11.56	0.00222
RC	2,872	4,146	7,018	48.25	0.00901
PR	<u>1,163</u>	<u>1,541</u>	<u>2,704</u>	<u>18.66</u>	<u>0.00337</u>
PPR	1,975	2,130	4,105	20.75	0.00475
PENS	4,246	4,989	9,235	34.13	0.01104
PPENS	5,020	5,422	10,442	36.00	0.01210
DEB	3,227	4,444	7,670	50.31	0.00969
<i>Toys and Games (n=100)</i>					
SA	542	955	1,497	11.93	0.00205
RC	2,194	3,454	5,648	37.37	0.00746
PR	<u>906</u>	<u>1,401</u>	<u>2,307</u>	<u>15.93</u>	<u>0.00303</u>
PPR	1,536	2,023	3,560	18.43	0.00443
PENS	3,339	4,725	8,064	30.11	0.01028
PPENS	4,037	5,062	9,099	33.39	0.01113
DEB	2,485	3,965	6,450	48.70	0.00855